vasic-digital tmux — CLAUDE.md

INHERITED FROM constitution/CLAUDE.md

All rules in `constitution/CLAUDE.md` and the `constitution/Constitution.md` it references apply **unconditionally** to this project. Project-specific rules below **extend** them — they do NOT weaken or override any universal clause. When this file disagrees with the constitution submodule, **the constitution wins**.

@constitution/CLAUDE.md

MANDATORY ANTI-BLUFF END-USER-QUALITY COVENANT

Forensic anchor — verbatim user mandate (2026-04-28, reasserted 2026-05-21):

> "We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Operative rule. The bar for shipping is **not** "tests pass" but **"end users can actually use the feature."** Every PASS — test OR HelixQA Challenge — MUST carry positive captured runtime evidence that the feature works for the end user. Metadata-only PASS, configuration- only PASS, "absence-of-error" PASS, and grep-without-runtime PASS are all critical defects regardless of how green the summary line looks. Tests and Challenges are bound EQUALLY.

This covenant is restated verbatim in every governance file at the consumer layer (`CLAUDE.md`, `AGENTS.md`, `QWEN.md`) so any tool that does not expand `@imports` still reads it. The same verbatim block lives upstream at `constitution/Constitution.md` §11.4, `constitution/CLAUDE.md`, and `constitution/AGENTS.md`. Removing or weakening this block is a §11.4-class release blocker.

Canonical authority: `constitution/Constitution.md` §11.4 and its sub-anchors §11.4.1 through §11.4.78. Project anchor: `Constitution.md` §101.

Non-compliance is a release blocker regardless of context.

> **Read first on a fresh conversation:** `CONTINUATION.md` (§0/§3/§8 are mandatory), then `Issues.md` for OPEN/PARTIAL/BLOCKED/RUNNING items. Canonical project authority: `Constitution.md` (Project Articles §101–§109) which extends `constitution/Constitution.md`.

Project overview

Verified hardened tmux 3.6a build with jemalloc, OOM protection, and per-session isolation via the `tmx` wrapper. Built around a hard verification gate that refuses to expose the binary unless functional tests pass. **Native dual-OS** (since 2026-05-13): runs as a host process on Linux AND macOS — no VM in the daily-use path. Each `tmx new -s NAME` creates its own tmux server with OS-native isolation: cgroup-v2 transient scope (`systemd-run --user --scope`) on Linux; POSIX rlimit wrapper (`RLIMIT_CPU` + `RLIMIT_NPROC`) on macOS. The session shell is the operator's host shell with full `$PATH`. Honest gap: `RLIMIT_AS` (memory) is NOT enforced by XNU for unprivileged processes — see `docs/guide/README.md` §5.6.

Critical base rules restated (for sessions that don't expand @imports)

- **No bluffing.** Every PASS — test OR HelixQA Challenge — carries positive runtime evidence (cgroup/`/proc` file *content*, `capture-pane` output, kernel log lines), never just an exit code. Universal §11.4 / Project §101.
- **FAIL-bluffs equally forbidden.** A test that exits FAIL for a script bug (not a product defect) is fixed at the source layer, never at call sites. Universal §11.4.1.
- **No-guessing.** Never use `likely`/`probably`/`maybe`/`seems`/`appears` in cause descriptions — prove with forensic evidence or mark `UNCONFIRMED:` / `PENDING_FORENSICS:`. Universal §11.4.6.
- **Operator-path coverage.** Every gate test exercises the SAME entry point an end-user invokes (`tmx new -s X`, not hand-spawned equivalents). Project §102.
- **Four-layer coverage.** Every defect lands all four layers before the cycle closes. Project §103.
- **CONTINUATION.md** updated in the SAME commit as any non-trivial state change. Universal §12.10.
- **commit_all.sh only.** Never `git push` directly on the main repo.

Commands (exact)

Step	Command
Install build deps (one-time; Linux needs root, macOS uses brew)	`bash scripts/install_deps.sh`
Full pipeline	`bash scripts/setup.sh`
Build only	`bash scripts/setup.sh --build-only`
Verify only	`bash scripts/setup.sh --verify-only`
Verification gate	`bash scripts/verify.sh`
Run all tests	`bash scripts/tests/run_all.sh`
Single test	`bash scripts/tests/NN_*.sh` (zero-padded, 01..17)

Step	Command
Constitution inheritance gate	`bash scripts/tests/test_constitution_inheritance.sh`
Meta-test (paired mutation)	`bash scripts/tests/meta_test_false_positive_proof.sh`
Containerized test run (bounded subset)	`bash scripts/test_containerized.sh`
VM test run (full suite)	`bash scripts/test_vm.sh` (`TMX_TEST_DESTRUCTIVE=1` for tests 12/13/14; `META=1` for meta-test)
End-to-end automation	`bash scripts/test_e2e.sh`
Commit + push (github+gitlab)	`bash commit_all.sh "message"`
Per-session wrapper	`tmx {new|attach|ls|kill}`

Never `git push` directly — use `commit_all.sh`.

Structure

Path	Role
`constitution/`	HelixConstitution submodule — universal governance (pinned `7f738df`) — **do not modify**
`tmux/`	upstream submodule (tag `3.6a`) — **do not modify**
`Containers/`	vasic-digital cgroup helpers submodule
`tmux/build*/`	build output (`.gitignore`'d)
`scripts/`	build, verify, install, 17 tests + inheritance gate, challenges, wrapper + conf templates
`scripts/tmx`	generated dispatcher (`.gitignore`'d) — from `tmx.template` (Linux) or `tmx-mac.template` (macOS bridge)
`scripts/tmux.conf.template`	the tmux config template (scrollback, copy-mode, Claude-Code TUI passthrough, clipboard)
`scripts/tests/meta_test_false_positive_proof.sh`	§103 layer-4 paired-mutation harness
`scripts/challenges/tmux.yaml`	HelixQA Challenge specs
`commit_all.sh`	only allowed push mechanism
`Constitution.md`	project authority — Project Articles §101–§109 (extends `constitution/`)

Path	Role
`Issues.md`	OPEN / PARTIAL / BLOCKED / RUNNING only
`Fixed.md`	RESOLVED items with closure SHA + evidence
`CONTINUATION.md`	live handoff state (must stay fresh)

Project-specific MANDATORY constraints

Build / packaging

- `scripts/setup.sh` detects host OS and invokes the right build pipeline (Linux ELF / macOS Mach-O) — Project §108.
- `scripts/tmux.conf.template` is the source for the generated tmux config; `setup.sh` substitutes OS-specific placeholders (clipboard command). Never hand-edit the generated `~/.tmux.conf` / wrapper.

Test / verification

- **Test-interrupt-on-discovery** (§103): any defect found during a test cycle STOPS the cycle. Fix at root cause + land all four layers (pre-build gate / runtime test / HelixQA Challenge / paired mutation)
  - rebuild + repeat from the beginning.
- **Operator-path coverage** (§102): gate tests use `tmx new -s X`.
- **Topology dispatch** (§104): tests detect topology and SKIP-with- reason — never silently degrade.
- **Destructive tests** (tests 12/13/14): require `TMX_TEST_DESTRUCTIVE=1` — run only on dedicated test hosts.
- **Constitution inheritance** is itself gated: `scripts/tests/test_constitution_inheritance.sh` + the `CM-CONSTITUTION-INHERITANCE` paired mutation.

Deployment / release

- `bash commit_all.sh "<msg>"` is the ONLY push mechanism — pushes to `github` + `gitlab`.
- `VERSION` is the single source of truth (`version=` semver + `versionCode=` monotonic). Both bump together with a `CHANGELOG.md` entry on every release.

Project-specific Applied Fixes Reference

#	Fix	Key files
A13	Per-session cgroup isolation (OOM contained per scope)	`scripts/tmx.template`
A15		`scripts/tmux.conf.template`

#	Fix	Key files
	Status bar showed `claude.exe` — strip `.exe` from window name	
A16	tmux scrolling fix for Claude Code TUI + mobile/Termux	`scripts/tmux.conf.template`, `scripts/tests/17_*.sh`
A17	HelixConstitution submodule + verified governance inheritance	`constitution/`, `scripts/tests/test_constitution_inheritance.sh`

Full forensic detail: `Fixed.md`.

Files to never edit directly

- `scripts/tmx` — generated by `setup.sh` from `scripts/tmx.template`
- `tmux/build*/` — build output (in `.gitignore`)
- `tmux/` — upstream submodule pinned to tag `3.6a`
- `constitution/` — HelixConstitution submodule (governance authority)

Project overrides of universal rules

None. This project introduces no override of any universal clause.

§11.4.87–98 — newly-inherited anchors (constitution 6828ff2)

The constitution submodule was ff-merged to `6828ff2` (2026-05-28). Twelve new universal anchors §11.4.87 through §11.4.98 apply unconditionally per §11.4.35. Long-form bodies live in `constitution/Constitution.md`; the short-form covenant mirrors below are operative.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch

Inherited from `constitution/Constitution.md` §11.4.87. Agent work continues until ALL of: Issues.md has zero non-terminal entries, CONTINUATION.md §3 active-work empty, no in-flight subagent, no external dependency in-flight. Every closure carries §11.4.4(b) four-layer coverage + captured-evidence per §11.4.5/§11.4.69. Tests AND HelixQA Challenges bound equally per the §11.4 forensic anchor. **Canonical authority:** `constitution/Constitution.md` §11.4.87.

§11.4.88 — Background-push (commit-flock released immediately, push detached)

Inherited from `constitution/Constitution.md` §11.4.88. `commit_all.sh` MUST release the commit flock as soon as `git commit` returns 0; the push runs detached via `nohup ... &` + `disown`. Per-remote `.git/.push.<remote>.lock` serialises same-remote concurrent pushes but allows different-remote parallelism. The only escape hatch is `--sync-push` for §11.4.41 force-push paths. **Canonical authority:** `constitution/Constitution.md` §11.4.88.

§11.4.89 — Background test execution

Inherited from `constitution/Constitution.md` §11.4.89. Any test expected to exceed ~30 s (`run_all.sh`, `meta_test_false_positive_proof.sh`, `test_e2e.sh`, etc.) MUST spawn via `nohup ... > <log> 2>&1 &` + `disown`; main work stream returns to the §11.4.42 priority queue immediately. Hard-dependent next steps poll the exit-status file. **Canonical authority:** `constitution/Constitution.md` §11.4.89.

§11.4.90 — Obsolete status + per-item obsolescence audit

Inherited from `constitution/Constitution.md` §11.4.90. The §11.4.15 Status closed-set gains terminal value `Obsolete (→ Fixed.md)` with mandatory `**Obsolete-Details:**` audit-line (Since + Reason from a closed vocabulary + Superseding-item + Triple-check evidence). The §11.4.23 colorizer marks Obsolete rows light-gray + strikethrough. **Canonical authority:** `constitution/Constitution.md` §11.4.90.

§11.4.91 — Summary-doc clarity

Inherited from `constitution/Constitution.md` §11.4.91. Every summary one-liner MUST be self-contained — ≥ 6 words OR ≥ 40 chars naming SUBJECT + PROBLEM/GOAL. Forbidden anti-patterns: section labels (`Composes with`, etc.), bare metadata (`Critical`, `Bug`, `In progress`), §-letter alone. Generators extract from the H1/H2 heading line per §11.4.54 ATM-NNN convention. **Canonical authority:** `constitution/Constitution.md` §11.4.91.

§11.4.92 — Multi-pass change-evaluation discipline

Inherited from `constitution/Constitution.md` §11.4.92. Every non-trivial change passes 5 evaluation passes BEFORE commit: (1) main-task captured evidence; (2) regression blast radius; (3) cross-feature interaction; (4) deep research + CodeGraph queries; (5) anti-bluff confirmation. Trivial exemption applies only when zero source code is touched and the commit message cites it. **Canonical authority:** `constitution/Constitution.md` §11.4.92.

§11.4.93 — SQLite-backed single source of truth for workable items

Inherited from `constitution/Constitution.md` §11.4.93. Text trackers (Issues / Fixed / Summary / CONTINUATION) become derived views of an authoritative SQLite DB at `docs/workable_items.db` with a Go binary at `cmd/workable-items/` providing bidirectional `md-to-db` / `db-to-md` sync. Round-trip must be byte-identical within tolerated whitespace. `commit_all.sh` refuses to commit while the diff is non-empty. **Canonical authority:** `constitution/Constitution.md` §11.4.93.

§11.4.94 — Zero-idle priority-first parallel-by-default operating mode

Inherited from `constitution/Constitution.md` §11.4.94. Idle is permissible ONLY when every queued item is genuinely externally blocked, OR operator STOP, OR §12 host-safety demand. Before any wake/sleep the conductor MUST survey parallel-work feasibility per §11.4.42 / §11.4.72 / §11.4.87 and dispatch non-contending items via §11.4.20 / §11.4.70 subagent-driven default. **Canonical authority:** `constitution/Constitution.md` §11.4.94.

§11.4.95 — Workable-items SQLite DB TRACKED in git, NEVER gitignored

Inherited from `constitution/Constitution.md` §11.4.95. Amends §11.4.93: `docs/workable_items.db` is authoritative source data, NOT a build artefact — TRACKED in git, never gitignored. Every md-to-db mutation stages + commits + pushes the DB alongside the MD regen per §11.4.19 atomic-move + §2.1 multi-upstream push. **Canonical authority:** `constitution/Constitution.md` §11.4.95.

§11.4.96 — Safe-parallel-work-with-long-build catalogue

Inherited from `constitution/Constitution.md` §11.4.96. Classifies parallel work during long builds as SAFE (docs/scripts/tests/submodule pushes per §11.4.88, research, DB ops, subagent dispatch per §11.4.20) vs UNSAFE (destructive git ops on source tree, mass deletes on hot subtrees, host-safety breaches). Conductor consults the catalogue at every pause point. **Canonical authority:** `constitution/Constitution.md` §11.4.96.

§11.4.97 — Maximum-use-of-idle-time + progress-update cadence

Inherited from `constitution/Constitution.md` §11.4.97. Strengthens §11.4.87 + §11.4.94 + §11.4.96: any idle minute during which work could autonomously progress and is not genuinely blocked is a §11.4.97 violation. Progress-update cadence: emit a 1-line operator-facing update at every commit landed / subagent return / anchor landing / captured-evidence acquisition / milestone closure. **Canonical authority:** `constitution/Constitution.md` §11.4.97.

§11.4.98 — Full-Automation Anti-Bluff: live tests re-runnable without manual intervention

Inherited from `constitution/Constitution.md` §11.4.98. Closes the manual-intervention gap §11.4.85/§11.4.87/§11.4.89/§11.4.94 did not explicitly forbid: every test (unit / integration / e2e / Challenge / stress / chaos / live) MUST be fully self-driving end-to-end and emit PASS/FAIL/SKIP-with-reason without further human action after startup. One-time credential bootstrap OUTSIDE test execution is the only permissible exception. Re-runnability = PASS at `-count=3` consecutive automated invocations. **Canonical authority:** `constitution/Constitution.md` §11.4.98.

§11.4.99 — Latest-Source Documentation Cross-Reference Mandate

Inherited from `constitution/Constitution.md` §11.4.99 (commit `9e3bcc5` ff 2026-05-28). Before committing any operator-facing instruction / guide / manual / setup walkthrough / troubleshooting cookbook, the author MUST fetch the LATEST official online docs of the third-party service / library being documented (WebFetch, MCP server, direct browsing — NOT training data) and cross-reference each instruction. A `## Sources verified <date>` section at the bottom of the document + a `Sources verified <date>: <urls>` footer in the commit message are MANDATORY. Re-verification cadence: every major release boundary; ≤ 6 months for general docs; ≤ 90 days for risk-classified families (messengers, cloud APIs, payments, LLM providers, code-hosting, OS/package managers). Stale docs graduate to §11.4.90 Obsolete after the 30-day grace. **Canonical authority:** `constitution/Constitution.md` §11.4.99 (commit `9e3bcc5`). Forensic anchor: Herald MTProto guide near-miss case study where stale guidance could have caused Telegram account ban.

§11.4.101 — Autonomous-decision-over-blocking mandate (User mandate, 2026-05-28)

Forensic anchor — verbatim user mandate (2026-05-28):

> "when working in endless working loop fully autonomously try to decide most properly about points which would block execution and wait for us. If we haven't answered now work would be blocked whole night! If possible and if that will not cause any issues make proper and most reliable and safe decision so we achieve maximal efficiency and work gets fully done!"

When operating in an autonomous / endless-loop mode (per §11.4.87), the agent MUST minimize operator-blocking and instead make the safe, reliable, reversible decision itself — so work is NOT stalled (e.g. overnight) waiting for input. §11.4.87 says keep working; §11.4.101 says HOW to clear the decision points that would otherwise force a stop-and-wait.

Decision rule (closed-set — proceed autonomously when ALL hold): (a) the action is reversible OR has a captured pre-op backup per §9.2; (b) the agent can determine the safe choice from captured evidence per §11.4.6 (no guessing — `LIKELY` is not a determination); (c) a wrong choice's blast radius is bounded AND recoverable; (d) it composes with anti-bluff §11.4, host-safety §12, data-safety §9.

Block-only-when rule (BLOCK via §11.4.66 ONLY when ALL hold): the action is irreversible AND high-blast-radius AND the safe choice cannot be determined from evidence — e.g. external-account state the agent cannot inspect, hardware it cannot access, destructive ops without backup, force-push (also §9.2 + §11.4.41), spending / sending to third parties. `Operator-blocked` per §11.4.21 is reached only after this rule fires AND the self-resolution-exhaustion audit completes.

Maximize-progress-while-blocked: an unavoidable block parks one work unit, it does not pause the loop — the agent MUST keep progressing every NON-blocked item in parallel per §11.4.87 + §11.4.94. Posing the question and going idle is a §11.4.94 + §11.4.97 violation.

Composes with §11.4.6 / §11.4.21 / §11.4.40 / §11.4.41 / §11.4.66 / §11.4.87 / §11.4.94 / §9.2 / §12. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-101-PROPAGATION` enforces the literal anchor `11.4.101` across the consumer fleet; paired §1.1 meta-test mutation strips the literal → gate FAILs (gate-code = separate work item). No escape hatch — no `--always-block-on-decision`, `--never-decide-autonomously`, `--skip-decision-rule`, `--block-without-self-resolution` flag exists.

Canonical authority: constitution submodule `Constitution.md` §11.4.101.

Non-compliance is a release blocker regardless of context.

§11.4.102 — Mandatory systematic-debugging activation + always-loaded skill-discovery + plugin-dependency availability (User mandate, 2026-05-29)

Forensic anchor — verbatim user mandate (2026-05-29):

> "Make sure that we ALWAYS trigger / start the "/superpowers:systematic-debugging" skills when any issues happen! If this is possible to activate and use in this situations out of the box when we spot problems / issues / bugs / misalignments / unconsistencies we MUST activate the skill(s) and make strongest efforts in full in depth analisys / debugging and determine root causes of all problem or obtain relevant data and information we need! ... we MUST make sure that "/using-superpowers" skill is ALWAYS loaded, applied and used! All dependencies (plugins) that Claude Code or other market places are offering MUST BE installed if these are not already available for loading and use!"

Three cooperating invariants — the difference between guess-and-retry and investigate-to-root-cause-first.

(A) Mandatory systematic-debugging activation. On ANY spotted issue / bug / test failure / gate failure / regression / misalignment / inconsistency / unexpected behaviour, the agent MUST activate `superpowers:systematic-debugging` (or the platform-equivalent structured-debugging discipline) **BEFORE proposing, writing, or applying any fix** — the **Iron Law: NO FIXES WITHOUT ROOT CAUSE INVESTIGATION FIRST.** Full four-phase arc: root-cause (read the real failure, reproduce, gather facts) → pattern (classify the defect, search for the same pattern elsewhere) → hypothesis (form a falsifiable cause, prove/disprove with captured evidence) → implementation (design the fix only against the proven root cause). Guess-and-retry, symptom-patching, and re-running a failed test hoping it passes

("probably transient / flaky") WITHOUT a completed investigation are §11.4.102 violations. Real-world signal: calling a failure `transient` / `flaky` / `intermittent` / `probably-timing` without captured forensic evidence is simultaneously a §11.4.6 (no-guessing) and §11.4.7 (demotion-evidence) violation — §11.4.102 makes the corrective response mechanical (the classification is forbidden until the systematic-debugging arc proves it).

(B) Mandatory always-loaded `using-superpowers`. `superpowers:using-superpowers` (or the platform-equivalent skill-discovery / capability-index discipline) MUST be loaded and applied at session start and consulted before any task. Operative rule: before acting on ANY request, survey available skills; if ANY skill could apply — even at 1% relevance — it MUST be invoked rather than improvised from memory. Skipping skill-discovery at session start, or improvising a task a loaded skill exists for, is a §11.4.102 violation.

(C) Mandatory plugin / dependency availability. Every skill plugin / marketplace package / capability dependency the project relies on (Claude Code, its skill marketplaces, or any other runtime's plugin ecosystem) MUST be installed + loadable BEFORE the dependent work proceeds. A missing plugin that blocks a mandated skill (e.g. `superpowers` absent so `systematic-debugging` cannot launch) is a **release-blocker** until installed + confirmed loadable. Install mechanism: the runtime's own plugin/marketplace install path — for Claude Code the in-session `/plugin` marketplace flow (add marketplace → install plugin → confirm the skill appears in the available-skills list); for other runtimes the documented package/ extension installer. Anti-bluff: confirm by observing the skill in the live capability list, never assume install succeeded (install exit 0 ≠ skill loadable, per the §11.4.80 lesson).

Composes with §11.4.4 (test-interrupt — first action after STOP is launch systematic-debugging), §11.4.6 (no-guessing — (A) is the procedural enforcement producing the facts §11.4.6 demands), §11.4.7 (demotion-evidence — the arc captures same-conditions evidence), §11.4.8 (deep-web-research — inside the hypothesis phase), §11.4.43 (TDD-fix — RED test written against proven root cause), §11.4.70 (subagent-driven — the investigation is a natural subagent dispatch), §11.4.82(A) (generalises Phase-1-forensic-before-speculative-patch to ANY fix + adds skill-discovery + plugin-availability), §11.4.92 (feeds Pass 1 + Pass 4). Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-102-PROPAGATION` (literal `11.4.102` across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no `--skip-systematic-debugging`, `--guess-and-retry-OK`, `--symptom-patch-permitted`, `--skip-skill-discovery`, `--plugin-optional`, `--missing-plugin-is-warning` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.102.

Non-compliance is a release blocker regardless of context.

§11.4.103 — Continuous parallel-stream working routine (User mandate, 2026-05-29)

Forensic anchor — verbatim user mandate (2026-05-29):

> "Do this working approach continuously and make it part of regular working routine and add it to the Constitution Submodule documented fully."

Promotes the proven multi-stream operating pattern into the project's **standing default working routine** (not a per-request opt-in). Binds §11.4.87/§11.4.88/§11.4.89/§11.4.94/§11.4.96 and adds the load-bearing invariant: **the main work stream MUST always stay FREE.**

(A) Main stream stays FREE. ALL commit AND push operations run detached (`nohup … &` + `disown`, per §11.4.88 commit-lock-release-immediately + detached-push) — the main stream returns to the priority queue the moment the local commit is durable, never blocking on a push or a slow mirror. **(B) ≥3 parallel background streams at all times + auto-backfill** (User mandate 2026-05-29; raised from ≥2) — run **at least three** subagent-driven background streams (per §11.4.70/§11.4.20, isolated per §11.4.58 PWU worktree + §11.4.84 quiescence) alongside the main stream whenever three-plus non-contending actionable items exist; **the moment any one stream is FULLY done, a new stream MUST immediately start and take its place** (claim next-highest-priority non-contending item), so the active-stream count NEVER drops below 3 while actionable items remain. Idle below 3 is permitted ONLY when no remaining non-contending actionable items OR all remaining are externally blocked (§11.4.94/§11.4.97/§11.4.101). Standing band 3–6, bounded above by §12.6 60% memory + §11.4.58 6-agent cap. **(C) Most-critical + most-visible first; audio always top** per §11.4.72 + §11.4.42 priority order. **(D) Safe-during-build scope only** — while the 42 GB containerised AOSP rebuild (§12.9) or any heavy `gradle/m -j` build runs, streams restrict to the §11.4.96 SAFE catalogue (investigation/forensic/docs/test-authoring/gate-authoring/read-only probes/submodule edits/research/DB ops/backgrounded pre-build+meta-test); NEVER a second concurrent heavy build (§12.8). **(E) Heavy anti-bluff on every closure** — root cause proven or `UNCONFIRMED:`/`UNKNOWN:`/`PENDING_FORENSICS:` per §11.4.6, captured evidence per §11.4.5+§11.4.69, deterministic consistency per §11.4.50, paired §1.1 mutations, §11.4.102 systematic-debugging on any spotted problem. **(F) Idle ONLY when genuinely externally blocked** (hardware/network upstream/in-flight build-test-push completion) OR operator STOP OR §12 host-safety, per §11.4.94(A)+§11.4.97; "nothing visible to do" with progressable items is NEVER valid; a block parks one work unit, not the loop (§11.4.101).

Composes with §11.4.58 / §11.4.70 / §11.4.72 / §11.4.87 / §11.4.88 / §11.4.89 / §11.4.94 / §11.4.96 / §11.4.97 / §11.4.101 / §11.4.102 / §11.4.42 / §11.4.84 / §12.6 / §12.7 / §12.8 / §12.9 / §9.2. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-103-PROPAGATION` (literal `11.4.103` across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no `--block-main-stream`, `--synchronous-commit`, `--synchronous-push`, `--single-stream-only`, `--skip-parallel-streams`, `--serialise-actionable-work`, `--idle-without-queue-survey` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.103.

Non-compliance is a release blocker regardless of context.

§11.4.104 — Participant identity, attribution & notification-tagging (User mandate, 2026-05-31)

Forensic anchor — verbatim user mandate (2026-05-31):

> "Every supported messenger must relate messages to participants (Subscribers/Users); the same logical person may have a different username on every messenger. Workable items must carry who created them and who they are assigned to. Notifications must @-tag the right participant — but never the operator (who drives the system) and never the system agent."

MANDATORY for every consumer that ships a messenger/notification surface (Herald + flavor binaries are the reference impl; others inherit per §11.4.35). Detailed spec: Herald `docs/design/PARTICIPANT_ATTRIBUTION.md` — restated, not redefined. **(A) Participant identity.** Every messenger MUST relate messages to a **Participant** (logical Subscriber/User); the SAME person MAY have a DIFFERENT username per messenger — modelled as a logical subscriber (canonical messenger-neutral `handle`, `kind ∈ {human,agent,service}`) + per-channel aliases (`channel`, `channel_user_id`, the `@username` used for tagging). Canonical handle closed set: `Claude` (reserved system-agent sentinel; never tagged) OR a subscriber `@username`. **(B) Operator = env var, not a DB flag** — the one human who drives via the agent CLI, designated by `HERALD_<CHANNEL>_OPERATOR_USERNAME` (e.g. `HERALD_TGRAM_OPERATOR_USERNAME`); a normal Participant whose handle equals that value. **(C) Workable items MUST carry `created_by` + `assigned_to`** (canonical handles): CLI-prompt→Operator; system/agent-detected→`Claude`; received-message→sender's resolved `@username`. `assigned_to` defaults to Operator, overridable. Legacy items carry `""` and MUST still parse + validate. **(D) Tagging matrix:** tag `assigned_to` if it is a human ≠ Operator; tag `created_by` if it is a human ≠ Operator ≠ `Claude`; NEVER tag `Claude` (system) or the Operator (no self-ping); de-dup; resolve each handle to the channel `@username`, skip if not on that channel. **(E) Anti-bluff (composes §11.4):** real SQLite round-trip with the new columns byte-identical (incl. legacy fixtures WITHOUT the fields); tagging matrix proven by a truth-table + a cell-flip mutation forcing FAIL; E2E real-event → real-message asserting the exact `@usernames` + a NEGATIVE case proving the Operator is NOT tagged; evidence under `docs/qa/<run-id>/`.

Composes with §11.4 + §11.4.1..§11.4.16 (anti-bluff covenant) / §11.4.5 / §11.4.69 / §11.4.50 / §11.4.91 / §11.4.93 / §11.4.95 / §1.1. Classification: universal (§11.4.17) — projects with no messenger surface inherit it latently (binds the moment they ship one, per the §11.4.96 pattern). Propagation gate `CM-COVENANT-114-104-PROPAGATION` (literal `11.4.104` across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no `--skip-attribution`, `--no-participant-tagging`, `--tag-operator-anyway`, `--attribution-later` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.104; detailed spec Herald `docs/design/PARTICIPANT_ATTRIBUTION.md`.

Non-compliance is a release blocker.

§11.4.105 — Natural-language intent recognition & clarification (User mandate, 2026-05-31)

Forensic anchor — verbatim user mandate (2026-05-31):

> "Users must NOT need to know command syntax (no `COMMAND: …` prefix). They send a clear natural-language message; the System determines the intent. The System recognizes the commands it has; if none matches it infers

> the exact intent; if it is totally unable it replies, tags the user (`@user …`), and asks to clarify precisely. We MUST always do our best to determine exact intent so we never annoy end users. This is a CORE part of the System."

MANDATORY for every consumer that ships a messenger/command surface (Herald + flavor binaries are the reference impl; others inherit per §11.4.35). Detailed spec: Herald `docs/design/INTENT_RECOGNITION.md` — restated, not redefined. **(A) No required command syntax.** Users MUST NOT be required to know any command syntax (no `COMMAND:` prefix); they send plain natural language and the System determines the intent. **(B) Three-tier resolution** (first that succeeds wins): TIER 1 — recognize the System's existing command set from natural language → action (confident deterministic match); TIER 2 — when no command matches, infer the exact intent (LLM dispatch maps language → action), NEVER guessing; TIER 3 — when neither a command nor a confident intent can be determined, REPLY to the message, TAG the sender (`@username`, resolved via the §11.4.104 IdentityResolver) and ask a PRECISE clarifying question NAMING the candidate intents — no guessing, no silent drop. **(C) Never guess, never drop:** a wrong action is worse than a clarifying question (composes §11.4.6 no-guessing); a message is NEVER silently dropped; only genuine ambiguity reaches Tier 3, which always replies-tags-and-asks. **(D) Anti-bluff (composes §11.4):** every tier ships unit + integration + E2E + full-automation tests with real captured evidence — Tier 1 truth-table (natural-language → action+fields, plus conservative negatives that MUST fall through to "no match"); Tier 3 E2E whose recorded reply body is EXACTLY `@<sender> <specific question>` + a NEGATIVE proving a clear command does NOT trigger clarify; a paired §1.1 mutation breaking the confidence guard (false-match) OR dropping the clarify tag MUST FAIL a test; evidence under `docs/qa/<run-id>/`.

Composes with §11.4 + §11.4.1..§11.4.16 (anti-bluff covenant) / §11.4.6 (no-guessing) / §11.4.104 (clarify reply tags the sender) / §11.4.5 / §11.4.69 / §11.4.98 / §1.1. Classification: universal (§11.4.17) — projects with no messenger surface inherit it latently (binds the moment they ship one, per the §11.4.96 pattern). Propagation gate `CM-COVENANT-114-105-PROPAGATION` (literal `11.4.105` across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no `--require-command-syntax`, `--guess-intent-ok`, `--skip-clarify`, `--drop-on-ambiguous` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.105; detailed spec Herald `docs/design/INTENT_RECOGNITION.md`.

Non-compliance is a release blocker.

§11.4.106 — Docs Chain — mechanical documentation/DB sync engine (Operator mandate, 2026-05-31)

Forensic anchor — operator mandate (2026-05-31): Docs Chain is the canonical mechanical enforcer of the documentation-sync mandates; consumers MUST use the engine instead of ad-hoc per-project doc-sync scripts, register their chains via per-context YAML, and never accept a faked transform.

MANDATORY for every consumer. Docs Chain (the `vasic-digital/docs_chain` engine — a universal Go bidirectional document-and-database dependency-propagation engine) is the canonical mechanical enforcer of the documentation-sync mandates. Detailed spec: the engine docs `~/Projects/docs_chain` → `docs/`

CONSTITUTION_INTEGRATION.md (distribution + inheritance + anchor mapping table) + docs/USE_CASE_CATALOGUE.md (chain recipes) — restated, not redefined. **(A) Use the engine, never ad-hoc scripts** — consumed **by reference** (the flat-layout sibling ~/Projects/docs_chain / the constitution-exposed path), inherited like §11.4.80's codegraph_* scripts: referenced, NEVER copied; ad-hoc sync_*/ generate_*_summary/update_readme_doc_links scripts are superseded and retired per registered context. **(B) Consumer-owned contexts** — the engine is project-agnostic; the consumer registers its chains as data via .docs_chain/ contexts/*.yaml (§11.4.28 decoupling); state.json + *.docs_chain.tmp are gitignored. **(C) Anchors it mechanizes** (per the CONSTITUTION_INTEGRATION mapping table): §11.4.12 / §11.4.53 / §11.4.45 / §11.4.56 / §11.4.57 / §11.4.59 / §11.4.60 / §11.4.65 / §11.4.86 / §11.4.93 / §11.4.95 / §12.10 / §11.4.44 — with content-hash change detection (NOT mtime, §11.4.86), atomic-rename + SQLite-txn commit + rollback (§9.2), both-dirty sync → conflict-not-silent-merge (§11.4.6), verify as the deterministic CI/pre-build gate (§11.4.50), per-run captured evidence to qa-results/docs_chain/<run-id>/ (§11.4.69). **(D) NOT a replacement for authoring discipline** — the source author still writes the §11.4.44 revision header; the engine only keeps exports in sync. **(E) Anti-bluff (composes §11.4):** the engine NEVER fakes a transform — a missing pandoc/weasyprint surfaces a typed ToolAbsentError + honest §11.4.3 SKIP-with-reason, never a fake PASS / partial write; every sync/verify carries real captured evidence.

Composes with §11.4 + §11.4.1..§11.4.16 (anti-bluff covenant) / §11.4.6 (no-guessing — conflict not silent merge) / §11.4.28 (engine/context decoupling) / §11.4.80 (inherited-by-reference, never copied) / §9.2 / §11.4.50 / §11.4.69 / §11.4.5 / §1.1, plus the sync anchors it mechanizes (§11.4.12/.53/.45/.56/.57/.59/.60/.65/.86/.93/.95/.44, §12.10). Classification: universal (§11.4.17) — projects with no derived-export/DB-sync surface inherit it latently (binds the moment they ship one, §11.4.96 pattern). Status (§11.4.6): engine Phases 1–3 AND the CLI/YAML loader (Phase 4) IMPLEMENTED+tested — cmd/ docs_chain/main.go registers sync/verify/diff(alias)/doctor, go test -count=1 ./... GREEN 7/7 packages; this CLI/loader is in the working tree but UNTRACKED and the engine is NOT yet a registered git submodule (in-tree at ./ docs_chain, HEAD = parent HEAD); only submodule distribution (Phase 6) remains PLANNED + OPERATOR-GATED — wire as phases land, claim no unshipped behaviour. Propagation gate CM-COVENANT-114-106-PROPAGATION (literal 11.4.106 across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no --ad-hoc-sync-ok, --skip-docs-chain, --fake-transform, --sync-evidence-optional flag.

Canonical authority: constitution submodule Constitution.md §11.4.106; detailed spec the Docs Chain engine docs docs/CONSTITUTION_INTEGRATION.md + docs/USE_CASE_CATALOGUE.md (vasic-digital/docs_chain).

Non-compliance is a release blocker.

§11.4.107 — Anti-bluff AV/test-validation techniques mandate (User-driven research, 2026-06-02)

Forensic anchor (genericised, 2026-06-02): a test PASSed on a SINGLE captured frame showing "a picture" on the target output — but the picture was a FROZEN / STALE frame from the previously-played content (stale-producer / stuck-decoder), so the feature was broken for the user while the test was green; a sibling incident

FLASHED media on the WRONG output for ~1 s before routing with no test sampling that window; a third class shipped a comparator that PASSed its own deliberately-degraded fixture (the analyzer, not the feature, was the bluff). §11.4.5 mandates captured evidence + a presence pass; §11.4.107 raises the bar to **liveness + correct-routing + self-validated-analyzer**.

Every test asserting audio/video output is genuinely playing/advancing for the end user MUST satisfy ALL of: (1) **a single captured frame is NOT proof** — prove LIVE, ADVANCING frames over a steady-state window via a **freeze-detection oracle** (near-duplicate / `freezedetect`-class filter OR perceptual-hash adjacent-frame distance, **NOT byte-identical compare** — byte-identity is only a zero-cost pre-filter); (2) an **independent frame-advance counter from the platform's compositor/decoder telemetry** must increase across the window (a different-domain oracle — flat counter ⇒ stuck decoder ⇒ FAIL even if pixels appear to move); (3) **loading/buffering is a distinct state** — wait for genuine playback-start before judging liveness, never false-FAIL a still-loading stream nor false-PASS a spinner; timeout+unreachable ⇒ SKIP-with-reason per §11.4.3, timeout+reachable ⇒ FAIL; (4) **not-stale-from-previous cross-check** — new content's first frame ≠ previous content's last frame; (5) **measured FPS / no-lost-frames within tolerance**; (6) **no-flash-on-the-wrong-output** — sample the non-target output at high frequency during a routing transition, any content frame there ⇒ FAIL (content-protection regime classified explicitly, never guessed); (7) **drive through the realistic feed/UI path, not deep-link shortcuts** (shortcuts bypass the transition paths where bugs live; UI-not-introspectable ⇒ operator-attended fallback per §11.4.52, never fake-PASS); (8) **metamorphic relations solve the oracle problem when there is no golden source** (same content on output-A vs output-B must match; paused ⇒ counter stops; 2× speed ⇒ ~2× advance rate); (9) **full-reference quality metrics (SSIM/VMAF/ΔE2000) vs a golden source for owned content**; (10) **mutation-test every analyzer with a golden-good + golden-bad fixture pair** so the analyzer itself provably cannot bluff (an analyzer that PASSes its golden-bad fixture is a bluff gate — §1.1 applied to the analyzers); (11) **per-channel audio RMS/loudness (EBU R128) + XRUN/underrun census** — a single aggregate RMS misses a dead channel; (12) **OCR overlay/subtitle detection needs a per-word confidence floor + ROI** to avoid BOTH false-positive (false-FAIL) and false-negative (false-PASS); (13) **thresholds calibrated on the project's own fixtures, not hardcoded from literature** (§11.4.6 no-guessing).

Honest gap (§11.4.6): true photon FPS at the sink has no clean software oracle — compositor/decoder counters measure the presentation pipeline; flag the gap, never claim it.

4-layer coverage per §11.4.4(b): pre-build gate (a `CM-AV-LIVENESS-NO-FROZEN-FRAME`-class gate asserting every output-is-playing test references the liveness battery not a single frame + an analyzer-self-validation gate wiring the golden-good/golden-bad fixtures into meta-test) + on-device/runtime test + paired §1.1 meta-test mutation (single-frame-only assertion → gate FAILs; analyzer that PASSes its golden-bad fixture → self-validation FAILs) + HelixQA Challenge. Every PASS via `ab_pass_with_evidence` citing the motion / not-stale / frame-advance / fps / per-channel-loudness / metamorphic / self-validation artefacts (`video_display` / `audio_output` / `subtitle_render` per §11.4.69) — never a single screenshot.

Classification: universal (§11.4.17) — platform-neutral AV/test-validation techniques reusable by ANY project validating media playback or any pixel/audio output; the project supplies its concrete capture mechanism + calibrated thresholds per §11.4.35. Composes with §11.4.5 (its strict expansion), §11.4.6, §11.4.50,

§11.4.68, §11.4.69, §11.4.85, plus §11.4.2 / §11.4.3 / §11.4.13 / §11.4.48 / §11.4.52 / §1.1. Propagation gate `CM-COVENANT-114-107-PROPAGATION` enforces the literal anchor `11.4.107` across the consumer fleet; paired §1.1 meta-test mutation strips the literal → gate FAILs (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.107.

Non-compliance is a release blocker regardless of context. No escape hatch — no `--single-frame-proves-playback`, `--skip-liveness`, `--byte-identical-freeze-OK`, `--no-frame-advance-counter`, `--skip-not-stale-check`, `--allow-wrong-output-flash`, `--deep-link-shortcut-OK`, `--unvalidated-analyzer-OK`, `--aggregate-rms-suffices`, `--hardcoded-thresholds-OK` flag exists.

§11.4.108 — Four-layer fix-verification + runtime-signature-as-definition-of-done mandate (systematic-debugging Phase 4.5, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): across one batch, multiple fixes were "green" at every gate yet NOT working for the end user — a change present in source and passed by both the source pre-build gate AND the post-build gate never reached the boot-time command-line embedded in the deployed boot artifact (output feature dead despite all-green); sibling fixes correctly built into the system image were masked by stale per-user overlay copies from a previous deployment (the running code was the stale shadow, not the fresh build); deployment did not wipe the mutable overlay so the shadow survived; validation ran against whatever was running — the stale shadow — and reported green on code that was never exercised. Per `superpowers:systematic-debugging` Phase 4.5: each fix revealing a fresh "fixed-but-not-working" in a DIFFERENT place is NOT independent bugs — it is ONE architectural VERIFICATION flaw; patching each symptom is thrashing.

A fix crosses FOUR distinct layers, and "fixed" at one does NOT imply fixed at the next: (1) **SOURCE** (committed in the source file — what a grep-the-source pre-build gate checks), (2) **ARTIFACT** (the change's BYTES actually landed in the produced build artifact — image / bundle / installer / embedded command-line), (3) **RUNTIME-ON-CLEAN-TARGET** (active on a CLEAN/fresh deployment — the layer the end user experiences — with no stale overlay shadowing the deployed code), (4) **USER-VISIBLE** (the feature works for the end user — the §11.4.5/ §11.4.69 captured-evidence layer). Green at layer 1 is the cheapest, least conclusive signal and says nothing about layers 2–4. A gate verifying ONLY the source layer is itself a §11.4 bluff surface.

The mandate (ALL must hold): (1) **Runtime-signature-as-definition-of-done** — a fix is DONE only when its declared runtime signature verifies on a CLEAN/fresh deployment; source-committed ≠ artifact-contains-it ≠ active-on-clean-target ≠ user-visible-working. (2) **Every fix declares ONE machine-checkable runtime signature** — a single observable on a clean target proving the fix is BOTH active AND working (a running-system property, a downstream/sink-side report per §11.4.13, a §11.4.5/§11.4.69 captured-evidence assertion, or a counter/state delta — NEVER a re-grep of the source); this **registry of per-fix runtime signatures is the SINGLE SOURCE OF TRUTH for "fixed"** — it REPLACES "a gate greps the source" as the definition of done. (3) **Gates span all four layers** — source (pre-build), artifact (post-build — assert the change's BYTES landed in the artifact, not merely that the source still contains the change), runtime-on-clean-target (post-

deploy — assert the runtime signature on a freshly-deployed clean target), user-visible (§11.4.5/§11.4.69). (4) **Eliminate the stale-deployment / shadow layer by construction** — deployment MUST yield a CLEAN state (wipe the mutable overlay) OR a pre-validation assertion MUST prove `running-artifact == built-artifact` BEFORE any validation runs; validation against possibly-stale deployed state is INVALID and any PASS it produces is a §11.4 PASS-bluff (the test exercised code that was never deployed). (5) **Meta-rule** — ≥ 3 "fixed-but-not-working" discoveries in one cycle signal an architectural VERIFICATION flaw, NOT three independent bugs; on the 3rd, STOP patching symptoms (§11.4.4), fix the VERIFICATION pipeline, and re-certify EVERY item through it on a clean target. (6) **A batch is "validated" only after COMPREHENSIVE per-item runtime-signature verification on a clean baseline** — NOT after the touched items' own tests pass.

Classification: universal (§11.4.17) — platform-neutral verification-pipeline disciplines reusable by ANY project that builds an artifact, deploys it, and validates the deployed result; the project supplies its concrete artifact format, clean-deployment / equal-artifact mechanism, and per-fix runtime-signature observables per §11.4.35. Composes with §11.4.1 / §11.4.2 / §11.4.4 (clause 5's STOP-and-fix-pipeline is its §11.4.108 specialisation; "clean baseline" = clause 4's clean deployment) / §11.4.5 / §11.4.6 (every layer asserted by evidence, never assumed to propagate) / §11.4.27 / §11.4.40 (§11.4.108 adds the cross-layer per-item runtime-signature dimension) / §11.4.46 (clean-baseline / equal-artifact pre-flight) / §11.4.50 / §11.4.52 / §11.4.69 (a runtime signature is a taxonomy-class observable) / §11.4.102 (Phase 4.5 architectural-flaw recognition IS clause 5's trigger). Propagation gate `CM-COVENANT-114-108-PROPAGATION` (literal `11.4.108` across the consumer fleet) + recommended per-fix gate `CM-RUNTIME-SIGNATURE-REGISTRY` + paired §1.1 meta-test mutations (strip the literal → propagation gate FAILs; downgrade a fix to source-only verification → registry gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.108.

Non-compliance is a release blocker regardless of context. No escape hatch — no `--source-green-is-done`, `--skip-artifact-byte-check`, `--validate-against-running-state`, `--no-clean-deployment`, `--skip-runtime-signature`, `--spot-validate-touched-only` flag exists.

§11.4.109 — Mandatory Anti-Forgetting Enforcement: PreToolUse Guard Hook + Subagent Constitutional Preamble + Orchestrator Pre-Action Checklist (Operator mandate)

Short tag: `anti-forgetting-enforcement`. **UNCONFIRMED forensic anchor** (pending operator's verbatim mandate quote). Background: emulator subagents ran raw host-direct `emulator`/`adb` because the orchestrator forgot to inject the Containers-submodule rule. A rule forgotten at dispatch is not enforcement. Fix: (A) a `PreToolUse` guard hook (`constitution/scripts/hooks/guard-forbidden-commands.sh`) that blocks host-direct emulator, force-push/bypass, sudo, and host-power commands at the tool-call boundary regardless of agent memory; (B) a canonical `docs/AGENT_GUARDRAILS.md` preamble the orchestrator pastes verbatim into every subagent dispatch; (C) an **ORCHESTRATOR PRE-ACTION CHECKLIST** in the same document. Hook = the floor; preamble = the ceiling.

Consuming projects MUST: (1) wire `constitution/scripts/hooks/guard-forbidden-commands.sh` as a `PreToolUse` hook in `.claude/settings.json` (or equivalent runtime settings); (2) maintain `docs/AGENT_GUARDRAILS.md` containing the `SUBAGENT CONSTITUTIONAL PREAMBLE` and `ORCHESTRATOR PRE-ACTION CHECKLIST` headings, with the anchor literal `11.4.109`; (3) provide a hermetic hook test suite (≥ 20 cases: every blocked class exits 2, every allowed command exits 0, escape hatch fires for non-power classes, host-power rejects even with escape marker). The hook is inherited by reference — NEVER copied locally (a copy diverges silently).

Gates: `CM-ANTI-FORGETTING-ENFORCEMENT` (hook present + wired + guardrails doc present + test present) + `CM-COVENANT-114-109-PROPAGATION` (anchor literal `11.4.109` across every consumer CLAUDE.md / AGENTS.md / QWEN.md). Paired §1.1 mutations: remove hook entry → gate (2) FAILs; delete guardrails doc → gate (3) FAILs; strip hook from constitution → gate (1) FAILs; strip `11.4.109` → propagation gate FAILs.

Classification: universal (§11.4.17). Composes with §11.4.6 / §11.4.10 / §11.4.75 / §11.4.76 / §11.4.78 / §11.4.79 / §11.4.80 / §11.4.81 / §11.4.84 / §11.4.98 / §11.4.102 / §12.

Canonical authority: constitution submodule `Constitution.md` §11.4.109; reference implementation `constitution/scripts/hooks/guard-forbidden-commands.sh` + `constitution/docs/AGENT_GUARDRAILS.md`.

Non-compliance is a release blocker. No escape hatch — no `--skip-pretooluse-hook`, `--no-guardrails-doc`, `--anti-forgetting-optional`, `--single-layer-sufficient` flag exists.

§11.4.110 — Pre-build build-readiness verdict + change-impact clash detection mandate (operator mandate, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): a fix shipped a new system-property *read* but introduced no matching security-policy grant + no property-context type entry — the read was silently denied at runtime + the feature was dead, while every pre-build gate stayed green because the gate only grepped the source file. The defect class generalises: a change introduces a new dependency on a *second* artifact (security policy, context file, service registry, interface freeze-snapshot, symbol table, build-graph node) that the pre-build never cross-checks. This is §11.4.108's SOURCE→ARTIFACT gap shifted LEFT to pre-build time — most such clashes are statically catchable from the change diff itself, before any build. The mandate (ALL hold): (1) a single deterministic **READY-FOR-BUILD verdict** gates the rebuild (orchestrator refuses to start the build unless READY); (2) a **diff-driven change-impact + clash detector** cross-checks every newly-introduced second-artifact dependency (new property read ⇄ property-context type + read-grant; new service ⇄ service-context entry; new init service ⇄ security label; new/changed stable interface ⇄ freeze-snapshot updated; new native-lib dep ⇄ module/prebuilt resolves; new policy rule ⇄ every type/attribute defined; two batch changes on the same seam ⇄ collision acknowledged); (3) **coverage-completeness is a gate** — every changed file maps to ≥1 gate + ≥1 deployed-target test + ≥1 paired §1.1 mutation, baseline ratchets upward per §11.4.50; (4) **two-speed honesty** — grep-speed always-on gates vs REQUIRES_BUILD heavy gates (build-graph parse-only dry-run, full neverallow compilation, ABI diff) as diff-gated opt-in stages, bounded per §12.6/§12.7; (5) every gate + wired analyzer is **anti-bluff by paired §1.1**

mutation; (6) **honest boundary** — a READY verdict proves static internal-consistency + ready-to-build, NOT that the feature works on the deployed target; the regime empties the *preventable* defect class, not the *all-defects* class (runtime/USER-VISIBLE remains §11.4.108's job).

Classification: universal (§11.4.17) — platform-neutral pre-build-rigor disciplines reusable by ANY project that builds an artifact from source; the consuming project supplies its concrete property/service/policy registries, build-graph parse-only command, interface-freeze mechanism, and changed-file→gate mapping per §11.4.35. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.9 / §11.4.27 / §11.4.50 / §11.4.67 / §11.4.75 / §11.4.92 / §11.4.108 (§11.4.110 is the SOURCE→ARTIFACT half shifted left to pre-build; §11.4.108 owns RUNTIME-ON-CLEAN-TARGET→USER-VISIBLE — together they span all four layers). Propagation gate `CM-COVENANT-114-110-PROPAGATION` (literal `11.4.110`) + recommended per-family gates `CM-READY-FOR-BUILD-VERDICT` / `CM-CHANGE-IMPACT-CLASH-DETECTOR` / `CM-COVERAGE-COMPLETENESS-GATE` / `CM-BUILDGRAPH-DRYRUN-WIRED` / `CM-SEPOLICY-NEVERALLOW-WIRED` + paired §1.1 mutations (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.110. Non-compliance is a release blocker. No escape hatch — no `--source-green-is-ready`, `--skip-clash-detector`, `--skip-coverage-gate`, `--no-ready-verdict`, `--grep-proves-neverallow`, `--skip-buildgraph-dryrun`, `--build-without-ready-verdict` flag.

§11.4.111 — Resolve-by-stable-name-not-by-enumeration-index mandate (research-derived, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): a platform bound an audio output to a kernel-enumerated device index (`card=0`); a second device of a different class enumerated FIRST at boot and took slot 0, shifting the intended device to slot 1 — the static index binding now pointed at the wrong device (policy mis-attached the sink, mis-assigned its TYPE, the output switcher labelled the AV receiver "Wired Headphone" + routing collapsed to stereo). The lower layer of the SAME stack already resolved the device correctly **by name** (scanning the controller-name registry) — proving the brittleness was the *index* binding, not the resolution capability. Generalises to every enumerated resource whose ordinal is assigned at discovery/boot/hotplug time (non-deterministic across reboots, device additions, topology changes). The mandate: any binding to a hardware device / resource handle / enumerated entity (audio cards, display connectors, network interfaces, storage devices, GPU render nodes, input/camera devices, container/process slots) MUST resolve by a **stable identifier** (name / UUID / serial / label / controller-name / content-hash / sink-reported identity) and MUST NOT bind by enumeration index / ordinal / slot, UNLESS the platform documents that ordinal as deterministically pinned AND the pin is itself captured + asserted as part of the binding. Where a stable identifier exists at one layer, every other layer binding the same resource MUST use the same identifier (mixed by-name-here / by-index-there is the structural weak link forbidden here). Honest boundary (§11.4.6): when only an ordinal exists, pin it deterministically via the platform's own mechanism, capture the pin in the binding's §11.4.108 runtime signature, and document the residual fragility as `UNCONFIRMED:`-class risk — never silently trust an unpinned ordinal.

Classification: universal (§11.4.17) — platform-neutral binding-robustness discipline reusable by ANY project binding to enumerated hardware/resources/handles; the consuming project supplies its stable-identifier mechanism (ALSA card name, DRM connector name, NIC predictable name, block-device UUID, etc.) + the layers that must agree per §11.4.35. Composes with §11.4.6 (no-guessing — "the index is *usually* stable" is the exact guess forbidden) / §11.4.8 (mature stacks resolve by name/UUID — reproducing a known-brittle index binding when the by-name path is documented is a §11.4.8 omission) / §11.4.69 (sink-side evidence verifies the by-name binding's correctness) / §11.4.108 (the by-name binding's runtime signature asserted on a clean target across the topology that broke the index) / §11.4.110 (a new ordinal binding in a diff is a statically-catchable clash class). Propagation gate `CM-COVENANT-114-111-PROPAGATION` (literal `11.4.111`) + recommended gate `CM-RESOLVE-BY-NAME-NOT-INDEX` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.111. Non-compliance is a release blocker. No escape hatch — no `--allow-index-binding`, `--ordinal-is-stable-enough`, `--skip-name-resolution`, `--trust-unpinned-index` flag.

§11.4.112 — Structural-impossibility won't-fix classification mandate (research-derived, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): deep research (§11.4.8) into relocating protected (content-protection / secure-surface) video to a secondary display PROVED — from authoritative platform/HDCP docs + reproducible captured behaviour (a secure surface is blanked on any output lacking the secure flag; mirror/screencap of a secure layer returns black) — that the goal is **structurally impossible by platform design**, not a missing feature or unsolved engineering problem. Without a durable classification, such a goal is re-investigated every cycle (re-read the same sources, re-run the same probes, re-derive the same impossibility — compounding wasted effort). The mandate: when deep research per §11.4.8 PROVES (cited authoritative sources AND, where applicable, reproducible captured evidence) a goal is structurally impossible on the target platform (forbidden by platform design / hardware-protocol constraint / documented kernel-or-API limitation — NOT merely unimplemented or hard), the goal MUST be: (1) classified `Won't-fix` + closed per §11.4.90 with closure reason `structurally-impossible`; (2) documented with the impossibility evidence — cited authoritative source URLs (per §11.4.99 latest-source verification) + the reproducible probe/captured evidence — in the tracker entry + relevant `docs/` guide; (3) NOT re-attempted in future cycles — a reopen MUST cite NEW evidence the platform constraint changed (per §11.4.34 + §11.4.7), never merely re-derive the same impossibility; (4) paired with the correct posture — the entry states what the project DOES instead, so "impossible" is never confused with "broken/unhandled". Honest boundary (§11.4.6): `structurally-impossible` is reserved for *proven* platform/hardware/protocol impossibility — "could not find a way" / "very hard" / "no time" are `Operator-blocked` (§11.4.21) or open work, NOT won't-fix; mislabelling them to avoid the work is a §11.4 planning-layer bluff. A future platform change can make the impossible possible; the classification is durable but not eternal.

Classification: universal (§11.4.17) — platform-neutral effort-conservation + honesty discipline reusable by ANY project; the consuming project supplies the specific impossible goal, its platform constraint, and the cited evidence per §11.4.35. Composes with §11.4.6 (FACT with cited evidence, never "probably can't") /

§11.4.7 (reopening requires NEW positive evidence) / §11.4.8 ("NO external solution found — structurally impossible" is the citation) / §11.4.34 (reopen attribution) / §11.4.90 (`structurally-impossible` is a closure reason in the closed vocabulary) / §11.4.99 (latest-source so the verdict is not stale). Propagation gate `CM-COVENANT-114-112-PROPAGATION` (literal `11.4.112`) + recommended gate `CM-WONT-FIX-STRUCTURAL-IMPOSSIBILITY` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.112. Non-compliance is a release blocker. No escape hatch — no `--wont-fix-without-proof`, `--reattempt-closed-impossible`, `--skip-impossibility-evidence`, `--impossible-equals-broken` flag.

§11.4.113 — Absolute no-force-push + merge-onto-latest-main mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03): "Any force-push is strictly forbidden! We must for every Submodule take as a base latest commit on Submodule's main (or master) branch, then on top of it carefully to merge all changes that have to be pushed! Once all merging is carefully done we perform commit and push to all Submodule's upstreams!"

Force-push is STRICTLY FORBIDDEN with NO exception — `git push --force`, `--force-with-lease`, `+<ref>`, or any history-rewriting overwrite of a remote ref, against EVERY repository this Constitution governs (main repo, this constitution submodule, every owned + nested submodule, every upstream). No operator-approval path, no "after a merge-first audit" path. The mandated 6-step integration procedure for any repo/submodule whose local has commits to publish OR whose mirrors diverged: (1) `git fetch --all --prune --tags` all remotes; (2) set the base to the LATEST commit on the canonical `main`/`master` branch (the most-advanced mirror tip); (3) carefully MERGE every change to be published on top of that base — union, preserve BOTH sides, NEVER `-s ours` / rebase / reset that drops commits (per §9 no-commit-loss); (4) resolve every conflict carefully — no conflict markers, no file dropped, gates/tests still pass; (5) commit the merge (stage only intended files, NEVER `git add -A` in a submodule per §11.4.30); (6) push to ALL upstreams — each push is a fast-forward because the merge commit descends from every mirror tip, so NO force is ever needed (if an upstream still rejects, return to step 1 for it, merge its new tip, re-validate, re-push). **TIGHTENS §11.4.41 / §11.4.71 / §9.2 / CONST-043 — the force-push escape hatch is REMOVED:** even WITH operator approval, even after a clean merge-first audit, force-push is forbidden, because the merge-onto-latest-main path is always available so force is never necessary. Those clauses' merge-first/fetch-first machinery stays in force as the integration discipline; only their terminal "...then force-push" step is struck.

Classification: universal (§11.4.17) — a platform-neutral integration discipline reusable across every repository. Composes with §2.1 (multi-upstream push — step 6 fans out) / §9 / §9.2 (absolute data safety — this is the no-loss push discipline that makes force unnecessary) / §11.4.4 / §11.4.6 (remote state unknowable without the step-1 fetch) / §11.4.26 (constitution-update conflict resolution IS this procedure) / §11.4.37 (fetch-before-edit → fetch-before-push) / §11.4.40 / §11.4.41 (merge-first stays, force-push step removed) / §11.4.71 (same tightening) / §11.4.88 (background-push still fast-forward-only) / CONST-043 (no force-push authorisable). Propagation gate `CM-COVENANT-114-113-PROPAGATION` (literal

11.4.113) + recommended gate CM-NO-FORCE-PUSH-ABSOLUTE (scan tracked scripts/hooks for push --force / --force-with-lease / push +<ref> + reject; a §11.4.109-class PreToolUse guard blocks the class at the tool-call boundary) + paired §1.1 mutation (inject a git push --force into a tracked script → gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.113. Non-compliance is a release blocker. No escape hatch — no --force, --force-with-lease, --allow-force-push, --force-push-authorised, --skip-merge-onto-main flag.

§11.4.114 — Last-known-good-tag regression isolation mandate (1.1.8-dev remediation, 2026-06-03)

When a previously-working feature/behaviour is observed broken, the FIRST diagnostic action MUST be to identify the last release tag (or commit/build/deploy) at which it was KNOWN-GOOD and diff/bisect the broken state against it — BEFORE any open-ended root-cause hunt or speculative fix. The known-good revision is the regression oracle: it bounds the search to the commits between good and now (git diff <good-tag>..HEAD --stat of the feature's files = captured evidence), tells you it is a regression REPAIR not a from-scratch design problem, and gives each suspect file a behavioural oracle. When the operator volunteers a known-good tag, that lead is load-bearing and MUST be acted on first. Default to a SURGICAL forward-fix (keep the post-good-tag features, revert ONLY the broken sub-part) over a wholesale revert that loses the batch's other working features — unless the operator prefers wholesale revert. Honest boundary (§11.4.6): "it worked before" is a HYPOTHESIS until the known-good tag is identified AND the feature is confirmed working there; "probably regressed in the last batch" without the diff is a guess; a feature that NEVER worked is not a regression. Composes §11.4.4 / §11.4.6 / §11.4.7 / §11.4.40 / §11.4.43 / §11.4.102 / §11.4.108. Propagation gate CM-COVENANT-114-114-PROPAGATION (literal 11.4.114) + recommended gate CM-REGRESSION-ISOLATED-AGAINST-KNOWN-GOOD + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.114. Non-compliance is a release blocker. No escape hatch — no --skip-known-good-diff, --root-cause-from-scratch, --assume-regression-source, --wholesale-revert-without-isolation flag.

§11.4.115 — RED-baseline-on-the-broken-artifact + polarity-switch mandate (1.1.8-dev remediation, 2026-06-03)

Strict refinement of §11.4.43's RED step. Every RED test MUST be authored to REPRODUCE the defect on the CURRENT, pre-fix artifact (the actual broken build/deployment), capturing positive evidence per §11.4.5 / §11.4.69 / §11.4.107 that the defect is genuinely present — never a synthetic failure the fix is then written to agree with. The SAME test source MUST carry a single polarity switch (env flag / parameter, canonical RED_MODE, default 1 = reproduce-and-assert-defect-present) that flipped to 0 post-fix converts the test into the GREEN regression-guard asserting the defect is ABSENT. One source, two roles: the bug-catcher IS the regression-guard; no separate happy-path test is authored as the primary guard (that demonstrates only that the test agrees with the fix — the §11.4.43 PASS-bluff). RED-on-broken-artifact then GREEN-on-fixed-artifact (on a clean target per

§11.4.108) MUST both be captured. Honest boundary (§11.4.6): if the RED run does NOT fail on the broken artifact, that is a finding (close per §11.4.7 with negative evidence, or fix the test) — a RED test that passes on the known-broken artifact is a blind test. Composes §11.4.1 / §11.4.2 / §11.4.5 / §11.4.69 / §11.4.107 / §11.4.4 / §11.4.7 / §11.4.43 / §11.4.50 / §11.4.108 / §11.4.114. Propagation gate `CM-COVENANT-114-115-PROPAGATION` (literal `11.4.115`) + recommended gate `CM-RED-POLARITY-SWITCH-PRESENT` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.115. Non-compliance is a release blocker. No escape hatch — no `--green-only-test`, `--skip-red-reproduction`, `--separate-happy-path-suffices`, `--synthetic-red-OK` flag.

§11.4.116 — Real-time conductor↔autonomous-test-framework sync channel mandate (1.1.8-dev remediation, 2026-06-03)

Any autonomous, long-running test/QA/validation framework an external orchestrator (conductor agent / operator) depends on for real-time decisions MUST expose a real-time sync channel: (1) a structured append-only event stream (JSONL or equivalent, one event per line, never rewritten) emitting at minimum session-start / phase-transition / per-test-or-challenge-start / captured-evidence-path / external-call (LLM / vision / sink-probe) / error / per-item-verdict events; (2) an atomically-rewritten status snapshot (single small file written write-temp-then-rename so a reader never sees a torn write) carrying current session/phase/item + counters + last verdict. Verdicts use the closed vocabulary PASS / FAIL / SKIP / OPERATOR-BLOCKED (§11.4.45). The conductor tails it live so it stays in real-time sync, can §11.4.4-interrupt on a fresh defect, and never idles blindly (§11.4.94 / §11.4.97). Anti-bluff: a verdict event MUST carry the evidence path that backs it (§11.4.69) — a PASS event with no evidence path is a channel-layer PASS-bluff; a snapshot reporting PASS while the stream shows no evidence event for that item is a contradiction → treat as FAIL; an item with no start-event cannot have a verdict-event. When the framework is an owned project-agnostic submodule (§11.4.28), the channel stays project-neutral — the consumer registers its data (endpoints, package ids, sink hosts) at runtime via the public API, never hardcoded. Composes §11.4.4 / §11.4.5 / §11.4.69 / §11.4.27 / §11.4.28 / §11.4.45 / §11.4.52 / §11.4.89 / §11.4.94 / §11.4.97. Propagation gate `CM-COVENANT-114-116-PROPAGATION` (literal `11.4.116`) + recommended gate `CM-AUTONOMOUS-FRAMEWORK-SYNC-CHANNEL` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.116. Non-compliance is a release blocker. No escape hatch — no `--no-sync-channel`, `--end-of-session-report-suffices`, `--verdict-without-evidence-path`, `--torn-status-write-OK` flag.

§11.4.117 — Computer-vision / OCR pixel-oracle fallback for non-introspectable UIs mandate (1.1.8-dev remediation, 2026-06-03)

Any test that needs to drive a UI control OR assert on-screen content MUST NOT assume the accessibility/semantic/DOM hierarchy is the source of truth for what the user sees. When the hierarchy is blank/partial/known-unreliable for the app under test (TV-Compose, leanback, canvas/`SurfaceView`/GL, games, custom-rendered UIs), the test MUST fall back to a PIXEL ORACLE: (1) DRIVE input by computer-

vision template-match (locate a control by its rendered appearance → tap its screen coordinates), not by a hierarchy node id; (2) ASSERT content by ROI OCR (read the rendered text the user reads — caption strip, title, error overlay) with a per-word confidence floor + region-of-interest per §11.4.107(12), not a hierarchy text attribute that may be absent/stale. The tool MUST both drive input AND read pixels — a hierarchy-only tool is NOT a content oracle. This makes §11.4.52's "near-empty hierarchy → INFEASIBLE" constructive: pixel-drive, not only SKIP. Anti-bluff (§11.4.107(10)): the CV/OCR analyzer is self-validated — golden-good fixture PASSes, golden-bad fixture FAILs, wired into meta-test; thresholds calibrated on the project's own frames, not hardcoded (§11.4.6). Honest boundary: when BOTH hierarchy is blank AND pixel oracle is infeasible (secure surface black-captures per §11.4.112, geo-unreachable), SKIP-with-reason per §11.4.3 + tracked operator-attended migration item per §11.4.52 — never fake PASS. Composes §11.4.5 / §11.4.6 / §11.4.48 / §11.4.49 / §11.4.52 / §11.4.107 / §11.4.112. Propagation gate `CM-COVENANT-114-117-PROPAGATION` (literal `11.4.117`) + recommended gate `CM-CV-OCR-PIXEL-ORACLE-FALLBACK` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.117. Non-compliance is a release blocker. No escape hatch — no `--hierarchy-is-content-oracle`, `--skip-pixel-fallback`, `--unvalidated-ocr-OK`, `--hardcoded-ocr-threshold-OK` flag.

§11.4.118 — Discovery-pressure to confirm known-issue-set completeness mandate (1.1.8-dev remediation, 2026-06-03)

A remediation/release cycle MUST NOT treat "every reported defect is fixed" as "the build is good" — the reported set is biased by what the operator happened to test ("we only see what we test"). After/alongside fixing the reported set, the cycle MUST run a discovery + stress pass across ALL target devices/environments that deliberately exercises subsystems, journeys, and edge cases BEYOND the reported defects — to CONFIRM the reported set is the COMPLETE critical set and surface unreported defects before the end user does. The pass MUST produce PROVABLE coverage: an enumerated list of the subsystems/user-journeys/stress scenarios actually exercised, each with its outcome (no-new-issue, or a new tracker entry per §11.4.15 / §11.4.16). "We found no other issues" is a §11.4 bluff unless accompanied by "here is the enumerated set we exercised" — absence of evidence of looking is not evidence of absence. New findings trigger §11.4.4 interrupt + the §11.4.114/ §11.4.115 isolation→RED→fix loop. Honest boundary (§11.4.6): discovery reduces the unknown-unknown surface but does not prove zero remaining defects — the earned claim is "reported set + enumerated discovery set addressed," with un-exercised subsystems stated as honest coverage gaps (§11.4.3), never silently implied clean. Composes §11.4.4 / §11.4.5 / §11.4.69 / §11.4.25 / §11.4.40 / §11.4.42 / §11.4.52 / §11.4.85 / §11.4.114 / §11.4.115 / §11.4.119. Propagation gate `CM-COVENANT-114-118-PROPAGATION` (literal `11.4.118`) + recommended gate `CM-DISCOVERY-COVERAGE-ENUMERATED` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.118. Non-compliance is a release blocker. No escape hatch — no `--reported-set-suffices`, `--skip-discovery-pass`, `--no-issues-without-coverage-list`, `--assume-complete` flag.

§11.4.119 — Single-resource-owner partitioning for parallel hardware testing mandate (1.1.8-dev remediation, 2026-06-03)

Strict refinement of §11.4.58 / §11.4.103 for hardware contention. When multiple parallel work/test/discovery streams exercise SHARED hardware or any exclusive-access resource (a media-playback path, a single HDMI/audio sink, an exclusive device handle, a serial/JTAG line, a GPU under exclusive capture), exactly ONE stream MUST own each such resource at a time. The exclusive owner drives it (playback, input injection, capture); every other concurrent stream targeting the same resource MUST be READ-ONLY (passive probes — `dumpsys` / `/proc` / `/sys` reads / sink-side network probes / log tails). Parallelism is partitioned by resource: distinct devices/sinks/handles run fully concurrent (stream-per-device), but the same device's exclusive resource is single-owner. Ownership MUST be enforced by an advisory lock/token (§11.4.58 L3 + the hardware analogue of §11.4.84 quiescence), event-driven (claim when the resource frees, release the moment work completes so the next queued stream claims it per §11.4.103 auto-backfill). Why: concurrent drivers of one exclusive resource produce CROSS-CONTAMINATED evidence (sink reports the wrong stream's audio, foreground belongs to whichever `am start` landed last, input events interleave) — a PASS under contention is a §11.4 evidence-integrity bluff. Honest boundary (§11.4.6): a "read-only" probe stream MUST issue NO state-changing command against a device it does not own; when a resource genuinely cannot be partitioned, streams serialize on it (single-owner over time), not run concurrently and hope. Composes §11.4.5 / §11.4.69 / §11.4.13 / §11.4.50 / §11.4.58 / §11.4.82 / §11.4.84 / §11.4.103 / §11.4.118. Propagation gate `CM-COVENANT-114-119-PROPAGATION` (literal `11.4.119`) + recommended gate `CM-SINGLE-RESOURCE-OWNER-PARTITION` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.119. Non-compliance is a release blocker. No escape hatch — no `--allow-concurrent-resource-drivers`, `--skip-ownership-lock`, `--read-only-may-mutate`, `--contended-evidence-OK` flag.

§11.4.120 — Fix-breaks-its-own-gate reconciliation mandate (1.1.8-dev remediation, 2026-06-03)

When a correct fix causes a pre-existing gate/test to FAIL because that gate asserted the OLD (now-removed-or-changed) behaviour, the gate FAIL is the CORRECT signal the fix landed — it MUST NOT be suppressed by either forbidden response: (1) FAKE-PASSING the gate (editing it to `always pass`, weakening its assertion to a tautology, or deleting it — a §11.4 gate-layer bluff + a §11.4.84 mutation-residue risk); (2) REVERTING the correct fix to satisfy the stale gate (re-introducing the defect). The required response is RECONCILIATION: rewrite the gate to assert the NEW mechanism the fix introduced, backed by captured evidence of the new correct behaviour, AND update its paired §1.1 mutation so the mutation breaks the NEW invariant. Discriminator vs bluffing: after reconciliation the gate + mutation still form a valid §1.1 pair (mutate the new invariant → gate FAILs); a bluffed gate's mutation no longer makes it FAIL (the assertion became a tautology). The reconciliation MUST be a visible, evidence-cited change, never a silent assertion-weakening. Honest boundary (§11.4.6): a post-fix gate FAIL is NOT automatically "stale gate, reconcile" — investigate per §11.4.102 first; the FAIL may be the gate correctly catching a REGRESSION the fix introduced, in which case the FIX is wrong, not the gate. Reconcile ONLY when investigation PROVES the gate asserted

old-correct-now-removed behaviour AND the new behaviour is the intended, evidence-confirmed mechanism. Composes §11.4.1 / §11.4.4 / §11.4.6 / §11.4.84 / §11.4.102 / §11.4.108 / §1.1. Propagation gate `CM-COVENANT-114-120-PROPAGATION` (literal `11.4.120`) + recommended gate `CM-GATE-RECONCILED-NOT-FAKE-PASSED` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.120. Non-compliance is a release blocker. No escape hatch — no `--fake-pass-stale-gate`, `--revert-fix-for-gate`, `--weaken-assertion-to-pass`, `--delete-failing-gate` flag.

§11.4.121 — No-commit-while-build-writes-tracked-artifacts mandate (1.1.8-dev remediation, 2026-06-03)

A commit (especially `git add -A` / any broad stage) MUST NOT run while a build/packaging/generation step is actively writing artifacts INTO tracked (version-controlled) directories — doing so races the writer and stages a PARTIAL or stale artifact (a §11.4.108 SOURCE→ARTIFACT integrity failure landed in version control). The commit MUST be deferred until the build step that writes tracked artifacts has COMPLETED, so the tree is quiescent at the artifact layer AND the committed artifacts are the FRESH, whole outputs (no stale pre-rebuild artifact, no half-written file). Before committing tracked build outputs, verify the writing step finished (process exit / completion marker / per-artifact mtime ≥ build-start) — a build still in flight writing tracked dirs is a HOLD on the commit, not a race to win. Build-output analogue of §11.4.84 (no commit while a mutation gate is in flight): both close the "commit captures transient non-final tree state" class at two write-sources. Where build outputs land OUTSIDE version control (gitignored `out/` / `dist/`), the race does not apply. Honest boundary (§11.4.6): "the build probably finished" is not "the build finished" — verify with a completion signal; committing source changes that DON'T touch the build's tracked-artifact directories is fine mid-build. The PROJECT-SPECIFIC instance (which tracked directory + which build steps write it) is recorded in the consuming project's own governance per §11.4.35. Composes §11.4.6 / §11.4.30 / §11.4.58 / §11.4.84 / §11.4.88 / §11.4.96 / §11.4.103 / §11.4.108. Propagation gate `CM-COVENANT-114-121-PROPAGATION` (literal `11.4.121`) + recommended gate `CM-NO-COMMIT-DURING-ARTIFACT-WRITE` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.121. Non-compliance is a release blocker. No escape hatch — no `--commit-during-build`, `--stage-partial-artifact-OK`, `--assume-build-finished`, `--skip-build-completion-check` flag.

§11.4.122 — No-silent-removal-of-existing-components-without-operator-confirmation mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03):

> "Never ever remove any application, system component or service from already existing codebase / System without interactively asked question to us! THIS IS MANDATORY RULE / CONSTRAINT!"

Forensic case study (FACT). During the 1.1.8-dev burn-down, two shipped capabilities — F2 (an Apple-TV-class application) and F4 (a Huawei HMS / Mobile-Services component) — were removed from the existing System WITHOUT first asking the operator; the operator reversed both. A removal the operator has to discover and reverse after the fact is a defect of the same severity class as a §11.4 PASS-bluff: the System silently lost a user-facing capability the operator never agreed to drop.

No application, system component, service, package, feature, driver, module, library, prebuilt asset — any already-existing end-user capability of the existing codebase / shipped System — may be removed (deleted, dropped from the package set, disabled-into-non-shipping, un-bundled, de-listed, or otherwise made unavailable to the end user) WITHOUT FIRST interactively asking the operator and receiving an EXPLICIT keep-or-remove decision. The question MUST be posed through the platform's interactive clarification mechanism per §11.4.66 (`AskUserQuestion` on Claude Code) — NEVER a free-text "should I remove X?" buried in narrative, NEVER a silent removal justified post-hoc, NEVER an autonomous removal decision. A silent removal is a **release blocker** regardless of how well-intentioned the rationale (deduplication, "it was broken anyway", geo-restricted, incompatible, superseded) — the operator decides, the agent asks.

What counts as a removal (non-exhaustive): deleting an app/APK/binary from the build's package set (`PRODUCT_PACKAGES` / `device.mk` / equivalent), removing a service from the init/boot/service-registry set, dropping a kernel module / driver / config from the shipping configuration, un-bundling a prebuilt asset, deleting a submodule or its shipped output, removing a feature flag that gated a live capability, or any edit whose NET EFFECT is "an end-user capability that shipped before no longer ships." Adding, replacing-with-operator-approved-equivalent, or fixing a capability is NOT a removal. When uncertain whether an edit constitutes a removal, treat it AS a removal and ask (per §11.4.6 no-guessing + §11.4.101 — removal of an existing user-facing capability is high-blast-radius and MUST be operator-confirmed, never autonomously decided). The tracked DROP path: ask → operator approves → mark the item `Obsolete (→ Fixed.md)` with `Obsolete-Details` reason `feature-removed` + an operator-approval citation (§11.4.90) → then remove; the removal never precedes the operator's yes.

Classification: universal (§11.4.17) — a platform-neutral discipline reusable by ANY project that ships a set of user-facing capabilities; the consuming project supplies its concrete capability-manifest paths per §11.4.35. Composes §11.4.66 / §11.4.101 / §11.4.90 / §11.4.112 / §11.4.6 / §11.4.40 / §11.4.42. Propagation gate `CM-COVENANT-114-122-PROPAGATION` (literal `11.4.122`) + recommended gate `CM-NO-SILENT-COMPONENT-REMOVAL` + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.122. Non-compliance is a release blocker. No escape hatch — no `--remove-without-asking`, `--silent-removal`, `--autonomous-removal-OK`, `--dedup-removal-exempt`, `--it-was-broken-anyway` flag.

§11.4.123 — Rock-solid-proof-or-deep-research mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03):

> "Every single reported issue MUST BE fully and 100% validated with rock solid proofs! Nothing can be considered fixed or completed without hard evidence! No false results or bluff(s) of any kind is allowed! If we are not sure on how to achieve full testing, validation and verification of something we MUST ALWAYS perform deep web research for all possible data (articles, documentation, guides, and other resources) and opensourced codebases which we can use to solve our problems and perform testing with validation and verification which produces rock-solid evidence(s) and leaves no space for false results or any kind of bluff!"

Forensic case study (FACT). In the 1.1.8-dev remediation the validation method for two feature classes was, at first, genuinely unclear: relocating a `FLAG_SECURE` secure surface to a secondary display (pixel capture returns black) and asserting on-screen content in non-introspectable streaming-app UIs (blank accessibility hierarchy). Rather than declaring them "untestable" or accepting a metadata-only PASS, the cycle performed deep web research (`docs/research/testing_frameworks_20260603/`) that yielded the CV/OCR/liveness/sink-probe oracle stack (now §11.4.107 + §11.4.112 + §11.4.117) — making rock-solid evidence possible where it had appeared impossible. "Unclear how to validate" is a research trigger, NEVER a bluff licence.

Every single reported issue, every fix, and every claimed completion MUST be fully and 100% validated with rock-solid CAPTURED proof per §11.4.5 / §11.4.69 / §11.4.107 before it may be marked fixed / implemented / completed (§11.4.33 closure vocabulary). Nothing may be considered fixed or complete without hard captured evidence — metadata-only / configuration-only / absence-of-error / grep-without-runtime PASS are all forbidden (§11.4 / §11.4.1); no false results, no bluff of any kind, at any layer.

The research-or-don't-bluff rule (the operative addition): when the agent is UNSURE how to fully test / validate / verify something — when no obvious evidence-producing method exists OR the candidate method would yield only metadata/config/absence-of-error evidence — it MUST ALWAYS first perform deep web research per §11.4.8 + §11.4.99 (official docs, articles, guides, vendor references, standards, issue trackers, reusable open-source codebases) to DISCOVER or BUILD a validation method that produces rock-solid evidence and leaves no space for a false result. Declaring something "untestable" / "not automatable" / accepting a metadata-only PASS WITHOUT first exhausting this deep-research path is itself a §11.4.123 violation — same severity class as a PASS-bluff. The research output (cited source URLs + the evidence-producing method, OR the literal "NO external solution found — original work" per §11.4.8) is the captured proof the path was exhausted. Only after that research genuinely fails may the item be classified `PENDING_FORENSICS:` / `Operator-blocked` (§11.4.21) / `structurally-impossible` won't-fix (§11.4.112) — with the cited research as the evidence the classification is earned, never a convenience.

Classification: universal (§11.4.17) — a platform-neutral discipline reusable by ANY project; the consuming project supplies its concrete capture mechanisms + research corpora per §11.4.35. Composes §11.4.5 / §11.4.6 / §11.4.8 / §11.4.52 / §11.4.69 / §11.4.99 / §11.4.107 / §11.4.118 / §11.4.21 / §11.4.112. Propagation gate `CM-COVENANT-114-123-PROPAGATION` (literal `11.4.123`) + recommended gate `CM-ROCK-SOLID-PROOF-OR-RESEARCH` + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.123. Non-compliance is a release blocker. No escape hatch — no `--metadata-pass-suffices`, `--skip-proof`, `--untestable-without-research`, `--config-only-closure-OK`, `--bluff-when-unsure` flag.

§11.4.124 — Dead/unwired-code investigate-before-remove mandate (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04):

> "Before removing any seemingly-dead (zero-importer / unwired) codebase, we MUST investigate via git history where/how it was originally used and how it became dead. Removal is permitted ONLY when we have captured PROOF it is genuinely no longer needed — and that removal MUST be its own separate commit with a proper descriptive message. If there is no such proof, the code MUST be investigated for where/how it should be wired in properly, and any missing or unwired tests MUST be added. We MUST ALWAYS be extra careful with any codebase removal."

"Zero importers / never called / unwired ⇒ dead ⇒ delete" is a GUESS (§11.4.6), never a finding — a "no references" result proves only *current* non-reference, not genuinely-unneeded. Before removing ANY seemingly-dead element (zero-importer / never-called / unwired function / method / type / file / module / package / asset / config / build target) the agent MUST FIRST investigate via git history (`git log --follow`, `git log -S/-G` pickaxe across all history, blame on the deleted call-site) and capture as FACT: (1) WHERE/HOW it was originally wired in, (2) WHEN/HOW it became dead — call-site deleted deliberately / by mistake (regression) / never-completed / refactored-unreachable, (3) whether "no references" is real OR a hidden reference the static tool cannot see (reflection / dynamic dispatch / build-tags / codegen / DI / plugin registry / FFI / config-driven wiring). The investigation output (cited commits + determination) is the captured evidence. **Removal is conditional:** permitted ONLY with captured PROOF the element is genuinely no longer needed; that removal MUST be its OWN SEPARATE COMMIT (independently reviewable + revertible, composes §11.4.84 quiescence + §11.4.92 multi-pass) with a descriptive message citing the git-history evidence — plus §11.4.122 operator-confirmation when the element is an end-user capability; the §11.4.90 tracked path marks it `Obsolete (→ Fixed.md)`. **No proof ⇒ do NOT delete:** investigate WHERE/HOW to wire it in properly (restore a mistakenly-deleted call-site per §11.4.114; finish never-completed wiring) AND add any missing / unwired tests (§11.4.27 / §11.4.43 / §11.4.115 — the missing test is part of why it drifted into apparent-deadness). **Extra-caution default:** when uncertain whether removal-proof is sufficient, default to NOT removing (investigate + wire + test) per §11.4.6 + §11.4.101 + §11.4.122; "probably dead" is never sufficient — the bar is captured proof. Classification: universal (§11.4.17) — the consuming project supplies its static-analysis / importer-graph tooling + hidden-reference mechanisms per §11.4.35. Composes §11.4.6 / §11.4.8 / §11.4.84 / §11.4.90 / §11.4.92 / §11.4.101 / §11.4.114 / §11.4.122 / §11.4.27 / §11.4.43 / §11.4.115. Propagation gate `CM-COVENANT-114-124-PROPAGATION` (literal `11.4.124`) + recommended gate `CM-DEAD-CODE-INVESTIGATE-BEFORE-REMOVE` (a net-deletion commit must be removal-only + cite the git-history investigation OR be part of a tracked Obsolete item) + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.124. Non-compliance is a release blocker. No escape hatch — no `--zero-importers-means-dead`, `--delete-unwired-on-sight`, `--skip-git-history-investigation`, `--remove-without-proof`, `--bundle-removal-with-other-work` flag.

§11.4.125 — Code-review-agent gate before pre-build + main build (mandatory multi-layer review) (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04):

> "After all fixes/changes/implementations are done, BEFORE running pre-build tests and the main build, dispatch code-review agent(s) that analyze all work done + all existing data/facts + the existing codebase + current git history to determine quality, safety, and whether the fixes/changes will REALLY work; they MUST validate and verify that every test covering the fixes/changes genuinely validates the work with NO chance of false results or bluff of any kind. Any finding MUST be fixed, polished, improved, and covered with additional tests before the build proceeds. Multiple strong layers of checks."

After all fixes / changes / implementations in a batch are done, and BEFORE running the pre-build test sweep AND the main (artifact) build (for ANY project), the agent MUST dispatch one or more dedicated code-review agent(s) (subagent-driven by default per §11.4.70/§11.4.20) performing a multi-layer review that: (1) analyzes ALL work done in the batch (every fix/change + its source diff + stated intent); (2) analyzes ALL existing data + facts (captured evidence per §11.4.5/§11.4.69/§11.4.107, tracker entries, prior findings, the §11.4.108 runtime-signature registry); (3) analyzes the existing codebase (blast radius per §11.4.92, cross-feature interaction, contract integrity of every dependency); (4) analyzes current git history (what each change touched, how it composes with concurrent/recent work, whether it reproduces a known-broken pattern per §11.4.114/§11.4.124); (5) determines quality + safety + will-it-REALLY-work (robust + not error-prone — no solve-A-create-B; no host/data/security regression; genuinely delivers the end-user-visible behaviour per §11.4/§107); (6) validates + verifies the tests covering the work — every covering test genuinely exercises the work-under-test and catches its negation, with ZERO chance of a false result or bluff (a test that PASSes on broken-for-the-user work, a metadata-only/config-only/absence-of-error/grep-without-runtime assertion, or a gate whose paired §1.1 mutation does not make it FAIL is a finding). Any finding (defect / error-prone change / safety risk / will-not-really-work / bluff-or-false-result-capable test / missing-coverage gap) MUST be fixed, polished, improved, and covered with additional tests (four-layer per §11.4.4(b), TDD-RED-first per §11.4.43/§11.4.115) BEFORE the pre-build sweep + main build proceed; the review iterates (re-review after each remediation) until no blocking findings remain. The review is itself anti-bluff (its conclusions are captured evidence per §11.4.5/§11.4.69; a rubber-stamp review of a defective batch = PASS-bluff). It is one of MULTIPLE STRONG LAYERS — complementing, never replacing, the §1 pre-build sweep, §11.4.92 multi-pass (author-side self-review; §11.4.125 adds the structurally-separated reviewer seam per §11.4.70), §11.4.108 four-layer fix-verification, §11.4.110 build-readiness verdict, and the post-build / runtime-on-clean-target / user-visible layers. Composes §11.4 / §11.4.1 / §11.4.4 / §11.4.6 / §11.4.40 / §11.4.43 / §11.4.50 / §11.4.70 / §11.4.20 / §11.4.92 / §11.4.102 / §11.4.107 / §11.4.108 / §11.4.110. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-125-PROPAGATION` (literal `11.4.125`) + recommended

gate `CM-CODE-REVIEW-GATE-BEFORE-BUILD` (build starts only with a fresh code-review-completed marker for the current batch, produced after the last fix + before the pre-build sweep + main build) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.125. Non-compliance is a release blocker. No escape hatch — no `--skip-code-review`, `--build-without-review`, `--no-review-gate`, `--review-optional`, `--trust-the-author` flag.

§11.4.126 — Default autonomous-loop working mode from first prompt (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04):

> "Make sure that you continue work in endless fully autonomous loop, do not stop until new fully validated and verified version (tag) is created and published (all submodules and main repo) or IN A CASE OF some other main stream work until it is fully completed with all side work streams and nothing else is left in our working queue! THIS MUST BE ALWAYS the default working mode without us asking you! We tend to achieve ABSOLUTE EFFICIENCY, with this and all other projects which will incorporate this MANDATORY RULE / CONSTRAINT!!! This way of (your) working will be ALWAYS applied / followed / executed / fully respected, as soon as we assign / send first request (prompt) in the session! This stops only if we explicitly say so or nothing is left to be done in current working scope (release that will come / upcoming version)!!! Any mimicking (imitation) of this behavior / rules / mandatory constraints, false results or any kind of bluff(s) is ABSOLUTELY FORBIDDEN!!!"

The endless fully-autonomous loop is the **DEFAULT working mode**, engaged automatically the moment the operator sends the FIRST request / prompt of a session — the operator MUST NOT have to ask for it, request it, restate it, or re-enable it per session. §11.4.87 framed the endless-loop covenant as an explicit-instruction opt-in ("continue in endless loop fully autonomously" or a semantically-equivalent phrasing); §11.4.126 is the **capstone** that promotes the same covenant to always-on: from the first prompt onward, every agent operates in the §11.4.87 loop discipline as the standing default, with §11.4.94 zero-idle, §11.4.97 maximum-idle-use, §11.4.101 autonomous-decision-over-blocking, and §11.4.103 continuous-parallel-stream all engaged by default — no per-session activation handshake. The continuation contract: the loop continues until ONE of two terminal conditions holds — (A) **Release scope** — a new, fully-validated-and-verified version (tag) is created AND published across all owned submodules AND the main repo to all configured remotes (per §2.1 multi-upstream push + §11.4.40 full-suite-retest-before-tag + §11.4.113 absolute-no-force-push merge-onto-latest-main); OR (B) **Non-release main-stream scope** — the main-stream goal is fully completed AND every side work stream is done AND the working queue holds nothing left for the current scope. Until (A) or (B) holds, the agent MUST keep working (claim the next priority item, dispatch the next parallel stream, progress every non-blocked item per §11.4.42 / §11.4.72 / §11.4.94 / §11.4.103). The loop STOPS ONLY on: (1) the operator explicitly saying so (STOP / pause / end); (2) nothing left to do in the current working scope — the upcoming release / current main-stream goal — with the queue genuinely empty per the (A)/(B) terminal conditions; (3) a §12 host-session-safety demand (the loop yields to host safety unconditionally). Idle-while-

blocked parks one work unit, it does not stop the loop — the agent keeps progressing every non-blocked item in parallel per §11.4.101 + §11.4.94 + §11.4.97. Goal — ABSOLUTE EFFICIENCY (no operator-side restart overhead, no idle gaps, no stop-and-wait round-trips); applies to this project AND every project that incorporates this Constitution. Anti-bluff: mimicking / imitating this loop behaviour, narrating continuation without performing it, fabricating progress, or emitting false / bluff results of ANY kind is ABSOLUTELY FORBIDDEN — this composes the entire §11.4 anti-bluff covenant family (§11.4 / §11.4.1 / §11.4.2 / §11.4.5 / §11.4.6 / §11.4.50 / §11.4.69 / §11.4.107); the agent MUST genuinely perform the continuous work and capture positive evidence for every closure, and a report claiming the loop ran while no real work / no captured evidence was produced is a §11.4 PASS-bluff at the operating-mode layer. Classification: universal (§11.4.17). Composes with §11.4.87 (the endless-loop covenant — §11.4.126 promotes it from opt-in to always-on default) / §11.4.94 / §11.4.97 / §11.4.101 / §11.4.103 / §11.4.66 / §11.4.6 / §11.4.40 / §11.4.42 / §11.4.72 / §11.4.113 / §2.1 / §12. Propagation gate `CM-COVENANT-114-126-PROPAGATION` (literal `11.4.126` across the consumer fleet) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.126. Non-compliance is a release blocker. No escape hatch — no `--ask-before-continuing`, `--single-turn-only`, `--not-default-loop`, `--mimic-OK` flag.

§11.4.127 — Session-handoff resumption-prompt mandate (User mandate, 2026-06-06)

Forensic anchor — verbatim user mandate (2026-06-06): "make sure that in situations like this now when new session is needed you ALWAYS prepera such sentence - which will be valid for particular moment and the phase of the project and enough for work to continue."

When the agent determines a fresh session is needed (context-window limits, performance degradation) OR the operator asks whether a new session is needed / requests a handoff, the agent MUST ALWAYS prepare + proactively provide a ready-to-paste **resumption prompt valid for that EXACT moment and project phase** — self-contained enough that pasting it into a fresh session resumes work with ZERO loss. Two variants on demand: a SHORT first-sentence ("Read `<handoff docs>`, then continue `<terminal goal>` …") AND a FULL detailed block. The prompt MUST: (1) point to the live handoff doc(s) — `.remember/remember.md` if present + `docs/CONTINUATION.md` per §12.10 — read FIRST + `git fetch --all`; (2) state current PHASE + immediate NEXT action + terminal goal; (3) embed exact live-state anchors (build IDs / artifact MD5, device/target serials, commit HEAD, in-flight PIDs + log paths, captured-evidence paths); (4) restate binding constraints (anti-bluff §11.4, no-force-push §11.4.113, exact version/naming, hardware/target gotchas); (5) be MOMENT-VALID, NEVER a generic template. Handoff doc(s) MUST be current BEFORE the prompt is given (§12.10). A missing / stale / generic prompt is a §11.4.127 violation. Composes §12.10 / §11.4.6 / §11.4.66 / §11.4.87 / §11.4.103 / §11.4.126. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-127-PROPAGATION` (literal `11.4.127`) + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.127. Non-compliance is a release blocker. No escape hatch — no `--skip-handoff-prompt`, `--generic-prompt-OK`, `--no-resumption-sentence`, `--handoff-without-state` flag.

§11.4.128 — Always-on device-recording mandate (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): we MUST ALWAYS live-record all available data from all devices we use for testing (or known to be under manual testing), EXTRA carefully so it never harms the device / its performance / causes side effects; raw recordings are NOT processed without need (token-conscious) and are ALWAYS git-ignored + code-intelligence-excluded; only curated evidence is committed, and only at release prep.

For EVERY test/debug device the project uses + every device under known manual testing, across EVERY reachable transport (USB / wireless ADB / SSH / serial / network introspection API), the project MUST ALWAYS live-record all analysable data: activities, all logs, performance metrics (CPU/memory/I/O/thermal/load), every sink-side report per §11.4.13, and any other live-changeable parameter. (1) **Extra-careful, side-effect-free** — non-invasive read-only probes only, bounded sampling, bounded write-volume, an observer-effect budget; a recorder that perturbs the device-under-test is a §11.4.128 violation, NOT evidence. (2) **Background + parallel + subagent-driven** per §11.4.103 + §11.4.70 — never blocks the main stream. (3) **Token-conscious — record-now, analyse-later** — raw data NOT processed without need; the only standing analyse-trigger is release-tag prep (§11.4.40 / §11.4.42) OR explicit operator ask. (4) **Raw is git-ignored (with a §11.4.77 regen-mechanism declaration) AND code-intelligence-excluded (§11.4.78/§11.4.79)** — only CURATED evidence is committed, and only at release prep under `docs/qa/<run-id>/` (§11.4.83). (5) **Deterministic layout** `<recording-root>/YYYY-MM-DD/<combined main+submodules state hash>/<DEVICE>_<SERIAL>/recording_NNN/<files>`. (6) **Anti-bluff** — a recorder claimed running but with no growing corpus is a §11.4 bluff; every curated finding traces to a real raw-corpus path; recorder health is itself captured evidence per §11.4.5/§11.4.69.

Composes §11.4.2 / §11.4.5 / §11.4.13 / §11.4.69 / §11.4.40 / §11.4.42 / §11.4.70 / §11.4.77 / §11.4.78 / §11.4.79 / §11.4.83 / §11.4.103 / §11.4.119. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-128-PROPAGATION` (literal `11.4.128`) + recommended gate `CM-DEVICE-RECORDING-ALWAYS-ON` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.128. Non-compliance is a release blocker. No escape hatch — no `--skip-recording`, `--record-without-layout`, `--commit-raw-corpus`, `--index-raw-corpus`, `--analyse-corpus-always`, `--invasive-probe-OK` flag.

§11.4.129 — Huge-blocker release protocol (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): when a huge blocker is discovered during release validation we MUST stop all testing, fix ALL discovered issues, process all recorded data from the last session, land rock-solid fixes, author

NEW validation+verification tests of ALL supported test types, rebuild, reflash, and RESTART the full validation+verification of every fix/change from the last release tag to now — on both devices in parallel, recorded, with real physical captured proofs and no bluff.

On discovery of a HUGE BLOCKER (release-blocking-severity defect: core user-facing capability broken, regression invalidating the in-flight cycle, or blast radius reaching the batch's other fixes) during release validation, execute in order with NO spot-check shortcut: (1) **STOP all testing** on every device (the §11.4.4 test-interrupt STOP at release granularity — continuing past a huge blocker is the §11.4 PASS-bluff). (2) **Fix ALL discovered issues** — not just the blocker; root-cause each per §11.4.102 + isolate regressions against the last known-good tag per §11.4.114. (3) **Process all recorded data from the last session** — analyse the §11.4.128 raw-corpus slice (this IS the §11.4.128(3) release-prep analyse-trigger). (4) **Land rock-solid fixes** per §11.4.123 + §11.4.43/§11.4.115 + §11.4.9. (5) **Author NEW validation+verification tests of ALL supported test types** per §11.4.27 + §11.4.85, each anti-bluff + paired §1.1 mutation. (6) **Rebuild (full, not module-only) + reflash to a CLEAN target** per §11.4.108. (7) **RESTART the full validation+verification from the last release tag to now** per §11.4.40 — RESTART, never resume — on both/all owned devices IN PARALLEL per §11.4.103/§11.4.119, every run RECORDED per §11.4.128, real physical captured proofs per §11.4.5/§11.4.69/§11.4.107, no bluff. This anchor BINDS the existing release anchors for the huge-blocker case (adds STOP→fix-all→process-recordings→new-tests-all-types→rebuild→reflash→full-restart + the restart-not-resume rule), citing them rather than duplicating.

Composes §11.4.4 / §11.4.40 / §11.4.42 / §11.4.9 / §11.4.27 / §11.4.85 / §11.4.102 / §11.4.108 / §11.4.114 / §11.4.115 / §11.4.123 / §11.4.128 / §11.4.103 / §11.4.119. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-129-PROPAGATION` (literal `11.4.129`) + recommended gate `CM-HUGE-BLOCKER-FULL-RESTART` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.129. Non-compliance is a release blocker. No escape hatch — no `--resume-after-blocker`, `--spot-validate-after-fix`, `--skip-recording-analysis`, `--skip-new-tests`, `--module-only-after-blocker`, `--single-device-restart` flag.

§11.4.130 — Post-remediation validate-the-fix-FIRST-after-redeploy (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): when a blocker discovered during release validation is fixed and a new artifact (rebuild / new flashing image / redeploy) is produced + the target reflashed, we MUST first re-test the SPECIFIC last-failing features + validate the just-incorporated fixes BEFORE the broader / full validation.

When a blocker / critical failure found during release validation is FIXED and a new artifact is produced + the target reflashed / redistributed / updated, the agent MUST: (1) **re-test the SPECIFIC last-failing features FIRST** (targeted guard tests for exactly the defects this fix addressed) BEFORE any broader / full-suite validation; (2) **validate the just-incorporated fixes with real captured evidence** — the §11.4.115 RED test flips GREEN at `RED_MODE=0` on the new artifact AND the §11.4.108 runtime-signature verifies on the CLEAN target the redeploy produced (metadata-only / config-only / absence-of-error / grep-without-runtime PASS

forbidden per §11.4 / §11.4.1; proof per §11.4.5/§11.4.69/§11.4.107/§11.4.123); (3) **only after the targeted fix is CONFIRMED working** proceed to the §11.4.40 full retest from the last tag to now. Rationale: a first fix attempt may not work / may be incomplete / may regress again under the new artifact — confirming the targeted fix FIRST catches a fix-did-not-take case immediately instead of hours later at the END of a full cycle (then restarting per §11.4.129); cheap-confirmation-first is §11.4.82 applied to the post-blocker reflash. This is the §11.4.46 recent-work-validation gate specialised for the post-blocker-reflash case + the targeted-confirmation phase that GATES §11.4.129's step-7 full-restart. Honest boundary (§11.4.6): "the fix probably took" ≠ "the fix took" — the RED→GREEN flip + runtime-signature on the new artifact is the proof; a still-FAILing targeted re-test re-enters the §11.4.114/§11.4.115 isolate→RED→fix loop, never proceeds to the full cycle on a still-broken fix. Composes §11.4.4 / §11.4.40 / §11.4.46 / §11.4.108 / §11.4.114 / §11.4.115 / §11.4.123 / §11.4.129 / §11.4.82. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-130-PROPAGATION` (literal `11.4.130`) + recommended gate `CM-FIX-FIRST-AFTER-REDEPLOY` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.130. Non-compliance is a release blocker. No escape hatch — no `--skip-targeted-retest`, `--full-cycle-first`, `--assume-fix-took`, `--validate-fix-at-end`, `--skip-red-green-flip-on-new-artifact` flag.

§11.4.131 — Standing session-resumption file mandate (User mandate, 2026-06-07)

Forensic anchor — verbatim user mandate (2026-06-07): "Make this markdown a standard file which will be written EVERY TIME when we need fresh session out of the box! It MUST BE always up to date and in sync so whenever new session is created all we have to do is just point to it!"

Every project MUST maintain a SINGLE canonical, always-current **session-resumption file** at a fixed, project-declared standard path (declared once per §11.4.35, never moved without a §11.4.66 operator decision). This file is the OUT-OF-THE-BOX entry point for any fresh session: creating a new session requires ONLY pointing the new agent at this one file. §11.4.131 promotes §11.4.127 (PREPARE a resumption prompt on demand) into a STANDING, version-controlled ARTIFACT — ALWAYS present, ALWAYS in sync. (A) **Existence + fixed path** — exists at the declared path at all times, encoded as a literal path in the project-layer instantiation (§11.4.35), never silently moved. (B) **Always written + always synced** — (re)written whenever a fresh session is needed OR the live state materially changes (new HEAD, build/artifact id, phase, device/target state, in-flight job, blocking decision) — the §12.10 trigger set; a stale resumption file is a §11.4.131 violation of the same severity class as a §12.10 stale-CONTINUATION violation. (C) **Content (composes §11.4.127)** — both SHORT + FULL variants; points to `.remember/remember.md` + `docs/CONTINUATION.md` read FIRST + `git fetch`; embeds exact live-state anchors (HEAD, build/artifact ids + checksums, device serials, in-flight PIDs + log paths, captured-evidence paths); states PHASE + immediate NEXT + terminal goal; restates binding constraints (anti-bluff §11.4, no-force-push §11.4.113, exact version/naming, hardware gotchas); MOMENT-VALID, never a generic template (§11.4.6). (D) **Export + freshness** — §11.4.65 scope (synchronized `.html`/`.pdf` siblings) + §11.4.44 revision header. (E) **Out-of-the-box resumption** — a fresh session, given ONLY this file's path, fully resumes with zero additional context. Composes §12.10 / §11.4.127 / §11.4.65 / §11.4.44 / §11.4.6 /

§11.4.66 / §11.4.126. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-131-PROPAGATION` (literal `11.4.131`) + recommended gate `CM-SESSION-RESUMPTION-FILE-PRESENT` + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.131. Non-compliance is a release blocker. No escape hatch — no `--skip-resumption-file`, `--ephemeral-prompt-only`, `--stale-resumption-OK`, `--generic-template-OK` flag.

§11.4.132 — Risk-ordered validation priority mandate (User mandate, 2026-06-07)

Forensic anchor — verbatim user mandate (2026-06-07): "We MUST ALWAYS first test and validate features, functionalities and fixes/changes that have been worked most recently, the ones which were most problematic, which have the most chance to crash or break again, the ones which have been re-opened the most times! Then, after we validate and verify all this with real (physical) proofs and hard evidence, with no false results and bluffs of any kind, we continue with all other existing tests in the test suites! This IS MANDATORY."

Tests / validations / verifications MUST run in **RISK-DESCENDING order** — the highest-risk set FIRST, and ONLY AFTER that set is fully GREEN with real (physical) captured evidence does the remainder of the suite run. Risk ranking is computed from a CLOSED set of factors, highest-risk first: (a) **most-recently-worked** features / fixes / changes; (b) **historically most-problematic** (longest defect history, most prior fixes/failures); (c) **highest crash/break/regress likelihood** (greatest blast radius / complexity / dependency surface); (d) **most-reopened** per §11.4.55 reopens-count (a high reopen count is the strongest empirical fragility signal). Each item in the highest-risk set MUST pass with real (physical) captured evidence per §11.4.5/§11.4.69/§11.4.107 — no metadata-only / config-only / absence-of-error / grep-without-runtime PASS (§11.4/§11.4.1), no false results, no bluff (§11.4.6). ONLY AFTER the entire highest-risk set is GREEN with captured proof does the rest of the suite run; running the suite in arbitrary order, or running lower-risk tests before the highest-risk set is GREEN, is a §11.4.132 violation. §11.4.132 REFINES/STRENGTHENS §11.4.130 (generalises "validate the just-fixed items first" to the full risk-ordered set) + §11.4.46 (adds explicit risk-ordering within the recent/high-risk set) + §11.4.42 (applies the implementation-layer priority discipline to VALIDATION ordering). Classification: universal (§11.4.17) — the consuming project supplies its recency / problematic-history / reopen-count sources (e.g. §11.4.93 workable-items DB `reopens_count`+`last_modified`) per §11.4.35. Composes §11.4.4/.5/.6/.7/.40/.42/.46/.50/.55/.69/.107/.130. Propagation gate `CM-COVENANT-114-132-PROPAGATION` (literal `11.4.132`) + recommended gate `CM-RISK-ORDERED-VALIDATION-PRIORITY` + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.132. Non-compliance is a release blocker. No escape hatch — no `--skip-risk-ordering`, `--any-order-OK`, `--suite-order-fixed` flag.

§11.4.133 — Target-System + hardware safety mandate (User mandate, 2026-06-08)

Forensic anchor — verbatim user mandate (2026-06-08): "Make sure that all changes we do to the System are ALWAYS safe for the System itself and for the hardware the system runs on! This is MANDATORY."

Every change to the TARGET system (firmware, kernel, init/boot scripts, drivers, sysfs/devfreq/voltage/clock/thermal/regulator register writes, partition/bootloader/U-Boot, HAL, framework, prebuilts, device config) MUST ALWAYS be safe for BOTH (a) the target System itself — MUST NOT brick, boot-loop, corrupt data, or render the device unrecoverable — AND (b) the hardware it runs on — MUST NOT exceed safe electrical/thermal/voltage/clock limits or damage panels/storage/radios/regulators. Concrete obligations: (1) reversible-first — verify irreversible high-blast-radius changes (bootloader/U-Boot MD5, partition layout) against known-good values + capture a pre-op backup (§9.2) BEFORE applying; (2) NO unverified hardware-control writes — never write an unverified value to a voltage/clock/regulator/thermal-throttle/current-limit sysfs node or register that could exceed datasheet limits, the safe range established as FACT (§11.4.6), never guessed; (3) thermal/perf changes (forcing a performance governor, pinning the top OPP, disabling thermal management) MUST respect the device's cooling design, validated by captured thermal evidence; (4) flashing MUST use the sanctioned tool + a freshly-built integrity-verified image — never an ad-hoc partition write or stale/unverified artifact; (5) unprovable-safety ⇒ blocked — a change whose target/hardware safety cannot be established from captured evidence is treated as UNSAFE and blocked (§11.4.6 + §11.4.101 reversible-first + §11.4.123 rock-solid-proof). DISTINCT from §12 host-session safety: §12 protects the DEVELOPER's HOST + session; §11.4.133 protects the TARGET device + its hardware — both apply, neither weakens the other. Classification: universal (§11.4.17) — the consuming project supplies its concrete hardware-control surfaces, datasheet-safe ranges, known-good bootloader/image hashes, and sanctioned flashing tool per §11.4.35. Composes §12 / §11.4.6 / §11.4.101 / §11.4.108 / §11.4.123. Propagation gate `CM-COVENANT-114-133-PROPAGATION` (literal `11.4.133`) + recommended gate `CM-TARGET-HARDWARE-SAFETY` + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.133. Non-compliance is a release blocker. No escape hatch — no `--unsafe-hardware-write`, `--skip-system-safety`, `--brick-risk-accepted` flag.

§11.4.134 — Code-review iterate-until-GO + rock-solid-evidence mandate (User mandate, 2026-06-08)

Forensic anchor — verbatim user mandate (2026-06-08): "For any fixes/changes given back to us for re-work by the code-review process, once we fix/improve everything per the code-review's requests, we MUST RE-RUN code-review AGAIN until we get a GO from it with NO new issues reported or warnings of any kind! All results produced by this whole process MUST ALWAYS give us rock-solid PHYSICAL evidence that the fixed/improved codebase really works now as expected, with no false results and no bluff(s) of any kind."

When the §11.4.125 code-review returns ANY finding — BLOCKING, nit, or warning — and the author fixes/improves the batch per that review, the code review MUST BE RE-RUN, and MUST KEEP being re-run after each remediation round,

until it returns a clean GO with ZERO new issues AND ZERO warnings of any kind. A single pass that "addressed the findings" is NOT sufficient: the corrected batch MUST pass a FRESH adversarial review (a re-review can surface NEW findings introduced by the very fixes that closed the prior ones — the §11.4.1 fix-A-creates-B failure mode). The loop terminates ONLY on a clean GO (no new findings, no warnings); a residual warning is itself a finding that re-arms the loop. Every round's verdict AND every fix's validation MUST carry rock-solid PHYSICAL captured evidence per §11.4.5 / §11.4.69 / §11.4.107 (captured audio / video / sysfs / dumpsys / sink-side / runtime-signature) proving the fixed/improved codebase REALLY works as expected — never metadata-only / configuration-only / absence-of-error / grep-without-runtime; no false results, no bluff at any round; a reported GO unbacked by captured physical evidence is itself a §11.4 PASS-bluff at the review-loop layer. §11.4.134 REFINES / STRENGTHENS §11.4.125 (iterate "until no blocking findings remain"): it makes the loop EXPLICIT (re-run after every remediation round, not once), raises termination to ZERO findings AND ZERO warnings (not merely zero-blocking), and BINDS rock-solid physical evidence to every round. Classification: universal (§11.4.17). Composes §11.4.125 / §11.4.1 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.69 / §11.4.107 / §11.4.50 / §11.4.108 / §11.4.123. Propagation gate `CM-COVENANT-114-134-PROPAGATION` (literal `11.4.134`) + recommended gate `CM-CODE-REVIEW-ITERATE-UNTIL-GO` + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.134. Non-compliance is a release blocker. No escape hatch — no `--skip-rereview`, `--single-review-pass`, `--warnings-ok`, `--evidence-optional` flag.

§11.4.135 — Standing regression-guard suite + every-fixed-defect-gets-a-permanent-regression-test (User mandate, 2026-06-08). Every project MUST maintain a STANDING regression-guard suite that runs on EVERY build+deploy and BLOCKS the release tag on any failure. Every closed defect (stable ticket id, e.g. ATM-NNN) MUST, in the SAME commit as its fix (extending the §11.4.43 DOCUMENT step), register a permanent §11.4.115 RED-on-broken-artifact regression test into the suite — `RED_MODE=1` capturing the historical defect on a pre-fix artifact (the proof the guard is real), `RED_MODE=0` the standing GREEN guard asserting the defect is ABSENT. A closure without a registered guard is a §11.4.123 violation. The suite runs FIRST in the post-deploy cycle (highest-risk set per §11.4.132) and is a §11.4.40 release-gate blocker. Forensic anchor (FACT): the wrong-subtitle-on-2nd-display defect was "fixed" via a source-side `CONTROL_MENU_LABEL_DENYLIST` that NO test mirrored or re-ran, so the NEXT chrome class recurred silently while the GREEN suite passed. Industry-standard bug-driven testing (Google content-driven testing; AOSP CTS/Tradefed) made mechanical + enforced. Composes §11.4.4 / §11.4.40 / §11.4.43 / §11.4.46 / §11.4.50 / §11.4.107 / §11.4.108 / §11.4.115 / §11.4.118 / §11.4.123 / §11.4.124 / §11.4.130 / §11.4.132. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-135-PROPAGATION` (literal `11.4.135`) + recommended gates `CM-REGRESSION-GUARD-REGISTERED` / `CM-REGRESSION-GUARD-SUITE-WIRED` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.135. Non-compliance is a release blocker. No escape hatch — no `--skip-regression-guard`, `--no-guard-on-close`, `--guard-optional` flag.

§11.4.136 — Real-content end-to-end playback-test mandate (User mandate, 2026-06-08). Refines/strengthens §11.4.107. Any test asserting media playback works MUST drive REAL content (catalog stream or offline reference clip) through the user's path (§11.4.48 UI-driven → §11.4.117 CV/OCR fallback) and assert it

genuinely PLAYS via the §11.4.107 liveness battery PLUS a decoder-health census — a numeric drop-buffer budget, no buffer-timestamp re-order/discard, no codec-reject (cite Android/Media3 ExoPlayer OEM pre-OTA playback-test mandate: "too many dropped buffers" >25, "unexpected presentation timestamp", "test timed out"). Metadata-only / launch-only / registration-only / single-frame / config-only PASS is forbidden (§11.4 / §11.4.1). A golden/reference clip corpus (BBC ExoPlayer testing samples) is the offline ground-truth. Composes §11.4.5 / §11.4.48 / §11.4.50 / §11.4.107 / §11.4.117 / §11.4.123 / §11.4.13 / §11.4.69. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-136-PROPAGATION` (literal `11.4.136`) + recommended gate `CM-REAL-CONTENT-PLAYBACK-TEST` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.136. Non-compliance is a release blocker. No escape hatch — no `--launch-proves-playback`, `--skip-decoder-health`, `--metadata-playback-pass-suffices` flag.

§11.4.137 — Subtitle/caption content-correctness oracle + secure-display-proxy-honesty mandate (User mandate, 2026-06-08). Refines §11.4.117 + §11.4.107 + §11.4.112. Forensic anchor (FACT): tests tasked to "physically verify the 2nd-display subtitle" PASSed GREEN while subtitles did NOT show / showed WRONG — the streaming player surface is FLAG_SECURE so `screencap -d <secondary>` returns BLACK (autonomous PIXEL verification structurally impossible per §11.4.112), so the test fell back to the accessibility-scraped/ `persist.atmosphere.subdebug` proxy, and the proxy accepted a chrome/menu LABEL (`Аудио и субтитры`) as a valid subtitle because the prose floor accepted any multibyte prose and NO menu-label denylist + NO position/cadence check existed. The mandate: a subtitle-correctness test MUST classify the cue's *content class* — a present cue is NOT a correct cue. CHROME (FAIL) if a known control/ menu label (closed multilingual deny-list MIRRORED from source, case-folded incl. non-ASCII), time/numeric chrome, not prose, OUTSIDE the lower safe-title band (CEA-708 9-anchor grid), OR STATIC across the window (real subtitle changes → ≥2 distinct prose cues, a metamorphic relation). DIALOGUE (PASS) only when prose + not-denied + not-chrome + position-ok + cadence ≥2 OR fuzzy-matches the SOURCE-extracted cue via normalized edit distance (§11.4.123 host ground truth). The oracle MUST be self-validated golden-good/golden-bad (§11.4.107(10)) and the deny-list MUST be verified present in the SHIPPED artifact (§11.4.108) — a source-green denylist with no test mirror + no artifact check is the exact recurrence pattern forbidden here. Secure-display honesty (§11.4.112): where FLAG_SECURE makes pixel verification impossible, the rock-solid autonomous proof is the player's caption telemetry + source-track presence + content-class oracle — NEVER a faked pixel "physical" pass; human-eye pixel confirmation is `operator_attended` (§11.4.52) with a tracked migration item. App-agnostic (keys off content class). Composes §11.4.3 / §11.4.5 / §11.4.6 / §11.4.107 / §11.4.108 / §11.4.112 / §11.4.115 / §11.4.117 / §11.4.123 / §11.4.13 / §11.4.69. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-137-PROPAGATION` (literal `11.4.137`) + recommended gate `CM-SUBTITLE-CONTENT-CORRECTNESS-ORACLE` + paired §1.1 mutation (strip the denylist/position/cadence check → golden-bad `Аудио и субтитры` PASSes → gate FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.137. Non-compliance is a release blocker. No escape hatch — no `--present-cue-is-correct`, `--skip-chrome-oracle`, `--length-heuristic-suffices`, `--pixel-pass-on-secure-display`, `--skip-position-check`, `--skip-cadence-check` flag.

§11.4.138 — Operator-escape => mandatory bluff-audit + permanent guard (User mandate, 2026-06-08). When the operator (or any out-of-band channel) finds a defect that the GREEN test suite passed, this is by definition a §11.4 PASS-bluff

— it MUST trigger, before the fix is closed: (1) a §11.4.102 systematic-debugging pass to FACT-root-cause; (2) a bluff-audit identifying the EXACT assertion that should have caught it but didn't, cited to `file:line` (canonical example: `lib/subtitle_content_validation.sh:sub_is_prose()` returning TRUE for `Аудио и субтитры`); (3) a permanent §11.4.135 regression guard registered in the SAME commit as the fix, with its §11.4.115 RED capturing the operator-found defect; (4) the bluff-audit committed under `docs/research/<scope>/<defect>_bluff_audit/`. Closing an operator-found defect WITHOUT the bluff-audit + permanent guard is itself a §11.4 violation (the bluff that let it through is still live and the defect will recur). Composes §11.4 / §11.4.1 / §11.4.102 / §11.4.108 / §11.4.115 / §11.4.118 / §11.4.123 / §11.4.135. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-138-PROPAGATION` (literal `11.4.138`) + recommended gate `CM-OPERATOR-ESCAPE-BLUFF-AUDIT` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.138. Non-compliance is a release blocker. No escape hatch — no `--close-without-bluff-audit`, `--operator-find-is-just-a-bug`, `--skip-permanent-guard` flag.

§11.4.139 — Fresh-process clean-artifact runtime-signature mandate (User mandate, 2026-06-08). Refines §11.4.108. Before any post-deploy validation — ESPECIALLY a non-pixel proxy verification (the subdebug/accessibility-cue channel used for FLAG_SECURE displays) — the harness MUST assert running-artifact == built-artifact: the deploy yielded a CLEAN target (mutable-overlay/userdata wiped) OR a pre-validation check proves no stale overlay shadows the deployed code (e.g. every guarded package — incl. the Presenter that emits the subtitle cue — resolves to the system partition, no per-user override). A stale shadow of the cue-emitting component (e.g. a Presenter APK predating the denylist) makes the proxy report on code that was never deployed — any PASS is a §11.4 PASS-bluff. Each fix declares ONE machine-checkable runtime signature verified on the clean target (the §11.4.108 registry IS the definition of done); for the subtitle class the signature is "the shipped Presenter APK contains the denylist literal (case-insensitive) AND the subdebug channel emits `candidate REJECTED reason=chrome-label` for a menu label." Composes §11.4.46 / §11.4.108 / §11.4.130 / §11.4.135 / §11.4.137. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-139-PROPAGATION` (literal `11.4.139`) + recommended gate `CM-CLEAN-ARTIFACT-RUNTIME-SIGNATURE` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.139. Non-compliance is a release blocker. No escape hatch — no `--validate-against-running-state`, `--skip-clean-precondition`, `--shadow-OK` flag.

> **§11.4.140 — Universal action-prefix system (`ACTION_NAME ::`) (User mandate, 2026-06-09; GRAMMAR_ADDENDUM 2026-06-09).** When a user prompt's FIRST non-blank line starts with a recognised action prefix, you MUST: (1) look the action token up in the action registry `constitution/actions/registry.yaml` (or `$HELIX_ACTION_REGISTRY`); (2) if it is a registered action, REPLACE the prefix with that action's `expansion` text and apply its `rules`; (3) execute the remainder of the prompt under the expanded instruction. **Four EQUIVALENT forms** — same action, same expansion, same execution: (1) `ACTION_NAME :: <rest>` (bare `::`), (2) `PREFIX::ACTION_NAME :: <rest>` (namespaced `::`), (3) `/ACTION_NAME <rest>` (bare slash), (4) `/PREFIX::ACTION_NAME <rest>` (namespaced slash). Thus `BACKGROUND :: x` ≡ `DEFAULT::BACKGROUND :: x` ≡ `/BACKGROUND x` ≡ `/DEFAULT::BACKGROUND x`. `PREFIX` is an action NAMESPACE; the reserved default namespace is **`DEFAULT`**, and an action runs WITH or WITHOUT the prefix. Grammar (all hold): anchored at the FIRST non-blank line only (mid-prose tokens never match); the action token

> AND the namespace are UPPERCASE-only `[A-Z][A-Z0-9_]*` (lowercase never matches); the namespace separator `::` inside the token carries NO surrounding spaces (`PREFIX::ACTION_NAME`), DISTINCT from the action-body separator `" :: "` (one ASCII space on each side of `::` — avoids C++ `Foo::Bar`, YAML `key: value`, URLs) in forms 1/2 and the slash-body separator (one space) in forms 3/4; stacked prefixes (`A :: B :: rest`) apply outer-to-inner, left-to-right (expand `A`, re-scan, expand `B`, then the residual is the task); a leading `\` escapes the prefix for BOTH the `::` and the slash form (`\BACKGROUND :: x`, `\/BACKGROUND x` — treat literally, strip the backslash, NO expansion) so action names can be discussed. **Conflict rule (slash form):** `/ACTION_NAME` (form 3) is honored as the action ONLY when `ACTION_NAME` (case-folded) does not collide with a built-in/host slash command (registry `slash_bare: auto` + `slash_conflicts: [..]`); form 4 (`/PREFIX::ACTION_NAME`) is ALWAYS unambiguous and always honored. An unknown token that matches the grammar shape (any of the 4 forms) but is NOT registered is NEVER silently expanded or silently dropped — ask which registered action was meant (§11.4.66 / §11.4.105) or treat it as a literal prompt, NEVER invent an expansion (§11.4.6); any prompt not satisfying the grammar is an ordinary prompt and the system is a no-op. The registered action **`BACKGROUND`** expands to: *"The following prompt that we will provide MUST BE executed in background in parallel with all main work streams using the subagents-driven development approach! All work done MUST PRODUCE rock solid evidence covered with hard physical proof(s) that all done is working as expected and as specified without any false results and without any bluff!"* (composes §11.4.20 / §11.4.70 subagent-driven, §11.4.58 / §11.4.103 parallel streams, §11.4.89 background execution, §11.4.5 / §11.4.69 / §11.4.107 captured physical evidence, §11.4 anti-bluff). The system is UNIVERSAL (every CLI agent reads this block via its context carrier per §11.4.35), extensible (new action = new registry row), decoupled + reusable (§11.4.28), and loads out-of-the-box. Classification: universal (§11.4.17). **Canonical authority:** constitution submodule `Constitution.md` §11.4.140. Non-compliance is a release blocker. No escape hatch — no `--skip-action-prefix`, `--ignore-prefix`, `--no-registry`, `--invent-expansion-OK`, `--single-layer-only` flag.

§11.4.141 — Token-efficiency mandate (research-derived + operator mandate, 2026-06-09). Every project worked on by AI coding agents MUST cut token spend (input AND output) toward **30–40% of current (a 60–70% reduction)** WITHOUT degrading quality/performance/safety or breaking any existing mechanism, via a composable, safety-ranked measure set: (1) **prompt-cache the static governance prefix** — the always-loaded governance forms a byte-stable cache-breakpointed prefix with no volatile bytes ahead of it; cache reads cost ~0.1× base input (the dominant cost driver — measured ~170K tokens of governance re-sent every turn, externally corroborated by Claude Code issue #24147); caching is transparent so it removes no rule, weakens no gate, changes no verdict — only billing (PRIMARY, biggest + safest lever); (2) **subagent model-tiering + output-to-file** — mechanical non-judgment work (search/grep/status/doc-export/read-only probes) to a Haiku-class model, the strong model RESERVED for all reasoning/verdicts/fix-design (§11.4.102)/code-review (§11.4.125)/demotion (§11.4.7), large output persisted to a file not an inline 350–520K-token transcript; the cheap model never emits a PASS so §11.4.50 + anti-bluff are untouched; (3) **thin always-loaded INDEX + on-demand detail** — concise index (one line per fix/anchor, EACH carrying the literal `11.4.N` token so propagation gates pass) with the canonical full text kept gate-scanned in `constitution/Constitution.md` and reachable in one hop — a de-

duplication realising §11.4.35, never a deletion; (4) **CodeGraph/retrieval-first over full-file loading** (§11.4.78/§11.4.79); (5) **output-token reduction** — terse status + `effort:"low"` on the mechanical allowlist only; (6) **tool-call batching + no re-reads**; (7) **compaction/context-editing for long sessions**. **Mandatory measured proof:** a token-accounting harness measures tokens-per-development-cycle BEFORE vs AFTER on a frozen deterministic workload from the authoritative `usage` object (input/cache_read/cache_creation/output split; NEVER `tiktoken`, NEVER the client-side cost estimate), reproduced N times (§11.4.50), pass = AFTER ≤ 40% of BEFORE OR the measured best-safe reduction with a cited cold-cache reason; the AFTER run MUST show ZERO regression on the pre-build sweep + meta-test mutation sweep + propagation gates + a strong-model reasoning probe + a cache-warm proof (`cache_read_input_tokens > 0`) — cost reduction with quality regression is a §11.4 FAIL. The headline number is the *measured* reduction, never the design estimate (§11.4.6/§11.4.123). No measure may break/degrade any existing mechanism, and the rule is structured so none can. Composes §11.4.5/.6/.20/.40/.50/.58/.69/.70/.78/.79/.80/.103/.106/.123/.125/.128/§12.6/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-141-PROPAGATION` (literal `11.4.141`) + recommended gate `CM-TOKEN-EFFICIENCY` + paired §1.1 mutation (inject a pre-breakpoint volatile token → cache collapses → measured reduction falls below bar → gate FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.141. **Project instantiation (§11.4.35):** [ATMOSphere consumer fills in: confirm the Claude Code governance-prefix cache is warm + free of pre-breakpoint volatiles; tier the §11.4.96-SAFE mechanical subagents (search/grep/status/doc-export) to Haiku with output-to-file under `qa-results/`; replace the consumer `CLAUDE.md` 112-row Applied-Fixes inline table + verbatim anchor restatements with a literal-anchor index pointing at `constitution/Constitution.md`; keep CodeGraph/lumen-first navigation; run the harness BEFORE/AFTER to prove the measured reduction.] Non-compliance is a release blocker. No escape hatch — no `--skip-token-efficiency`, `--no-cache-governance`, `--assert-reduction-without-measuring`, `--tier-down-reasoning`, `--inline-all-governance`, `--tiktoken-estimate-OK` flag.

§11.4.142 — Universal code-review mandate — every change reviewed, always, no exception (User mandate, 2026-06-09). Verbatim operator mandate: "ALL changes we do MUST pass through the code review step!!! ALWAYS!!!" EVERY change made to ANY repository this Constitution governs — without exception — MUST pass through an INDEPENDENT code-review step BEFORE it is accepted, committed, or built. NO change class is exempt: source code, fixes, tests, gates, meta-test mutations, documentation, doc-tooling, build/CI scripts, configuration, governance files (Constitution / CLAUDE / AGENTS / QWEN), conductor main-stream edits, sub-agent output, refactors, single-line edits — if a diff exists, it gets an independent review. This is the ABSOLUTE form of §11.4.125 (code-review agent gate after a batch, before pre-build sweep + main build): §11.4.142 strips every implicit scoping §11.4.125's "after all fixes/changes/implementations are done" phrasing could leave open — no "just a doc edit", no "just a one-liner", no "the author already self-reviewed (§11.4.92)", no "trivial change" carve-out. **Independence is load-bearing** — the reviewer MUST be structurally separated from the author (a dedicated code-review agent, subagent-driven by default per §11.4.70 / §11.4.20, or a distinct human), NEVER the author re-reading their own work; §11.4.92's multi-pass self-evaluation is the AUTHOR-side discipline and PRECEDES (never satisfies) §11.4.142. **The review is itself anti-bluff** (§11.4 / §11.4.1) — a rubber-stamp "looks good" is not a review; it MUST genuinely analyse correctness, safety (no host §12 / data §9 / target-hardware §11.4.133 regression), will-it-really-work (no solve-A-create-B), end-user behaviour (§11.4 / §107), and

test-genuineness (§1.1), its conclusions captured evidence per §11.4.5 / §11.4.69, and **it iterates to a clean GO per §11.4.134** — any finding (BLOCKING / nit / warning) re-arms the loop, acceptance only on ZERO new findings + ZERO warnings with rock-solid physical evidence. Honest boundary (§11.4.6): a passing review does NOT replace §11.4.108 four-layer runtime-signature verification nor the §11.4.40 full-suite retest — it is one of MULTIPLE STRONG LAYERS, and the FIRST one every change crosses. Composes §11.4.1 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.20 / §11.4.40 / §11.4.69 / §11.4.70 / §11.4.92 / §11.4.108 / §11.4.110 / §11.4.125 / §11.4.134 / §107 / §1.1. Classification: universal (§11.4.17) — the consuming project supplies its reviewer-dispatch mechanism + change-acceptance seam (commit wrapper / merge queue / PR gate) per §11.4.35. Propagation gate `CM-COVENANT-114-142-PROPAGATION` (literal `11.4.142`) + recommended gate `CM-EVERY-CHANGE-REVIEWED` (every accepted change carries a fresh independent-review-completed marker for its diff, produced by a reviewer distinct from the author, before acceptance/commit/build) + paired §1.1 mutation (strip the literal → propagation gate FAILs; accept a diff with no independent-review marker → `CM-EVERY-CHANGE-REVIEWED` FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.142. Non-compliance is a release blocker. No escape hatch — no `--skip-review`, `--no-review`, `--trivial-change-exempt`, `--doc-edit-exempt`, `--self-review-suffices`, `--review-after-commit` flag.

§11.4.143 — Real-user-journey mandate for video-streaming-app full-automation tests (User mandate, 2026-06-10). Verbatim operator mandate: "All video streaming apps ... require to choose some title and to press proper UI button to start or resume playing! Proper UI interaction to play exact show with proper content and subtitles is MANDATORY! Without it we just eventually play on 2nd display sample, and that's it mostly! THIS MUST BE ADDED as MANDATORY RULE regarding testing of any video streaming app in general with full automation tests! Universal in root constitution + ATMOSphere project extensions." Any full-automation test that asserts a video player / streaming application plays content MUST drive the REAL end-user journey through the app's OWN UI — launch → BROWSE the actual catalog → choose a SPECIFIC title → press the real Play/Resume button → confirm THAT chosen content is genuinely playing, with its correct subtitles, on the intended routing target. A test that bypasses the journey with a sample / demo / built-in-loop clip, a deep-link / `am start -a VIEW` / intent shortcut, a synthetic or pre-staged stream, or any path that does NOT exercise the app's own browse-select-play UI is a §11.4 PASS-bluff at the user-journey layer: it validates ROUTING (that *something* reaches the display) while leaving the user-visible behaviour the operator cares about — "the show I picked actually plays" — unproven (the operator's "we just eventually play on 2nd display sample, and that's it mostly" gap). The mandate (ALL hold): (1) **Real journey, not a shortcut** — launch → catalog browse → specific-title selection → real Play/Resume press through the app's own UI (§11.4.48 UI-driven), NEVER a deep-link / intent / sample / loop-clip shortcut; (2) **Chosen-content confirmation** — the PASS proves the SPECIFIC selected title is playing (not merely that pixels move on the target) via the §11.4.107 liveness battery on the §11.4.136 real-content path + the §11.4.137 subtitle content-correctness oracle for the chosen title's captions; (3) **Non-introspectable UIs use the pixel oracle** — when the app's accessibility hierarchy is blank/unreliable (TV-Compose / leanback / canvas / GL), DRIVE input + ASSERT content via the §11.4.117 CV/OCR pixel oracle, never a hierarchy-only tool; (4) **Login via the credential single-source** (§11.4.10), never hardcoded, never logged; (5) **Honest SKIP, never a faked PASS** — where the autonomous journey is genuinely infeasible (hard human-only login / CAPTCHA, geo-block per §11.4.3, secure-surface pixel-blanking per §11.4.112) the test is `operator_attended` SKIP-with-

reason per §11.4.52 + §11.4.3 with a tracked migration item — NEVER a metadata-only / sample-played / routing-only PASS. Honest boundary (§11.4.6): "the routing fired so the title is playing" is a guess — only the chosen-content liveness + subtitle oracle on the real journey proves it. Composes §11.4.48 / §11.4.107 / §11.4.117 / §11.4.136 / §11.4.137 / §11.4.52 / §11.4.3 / §107 / §1.1. Classification: universal (§11.4.17) — the consuming project supplies its concrete app roster, login-credential source, routing target, and UI-driving / pixel-oracle harness per §11.4.35. Propagation gate `CM-COVENANT-114-143-PROPAGATION` (literal `11.4.143`) + recommended gate `CM-VIDEO-REAL-JOURNEY-TEST` (every video-streaming-app playback test drives the real browse-select-play journey + confirms the chosen content + subtitles, or SKIPs-with-reason) + paired §1.1 mutation (replace a real-journey test's browse-select-play path with a deep-link / sample shortcut → gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.143. Non-compliance is a release blocker. No escape hatch — no `--sample-playback-ok`, `--skip-real-journey`, `--deep-link-suffices`, `--routing-only-pass`, `--no-title-selection` flag.

§11.4.144 — Tracked/recorded-device availability-following mandate (User mandate, 2026-06-10). Direct operator mandate: every device the project is tracking / following / recording (test / debug / manual-testing device, across every reachable transport — USB / wireless ADB / SSH / serial / network introspection API) MUST be availability-FOLLOWED — its connection state continuously monitored, any drop handled, never silently abandoned and never presented as a continuous recording. The §11.4.128 always-on recorder KNOWS a tracked device is absent (its per-device loop guards on the reachable state) but, lacking a following discipline, merely spins idle — no data captured, no offline event logged, no resume, no escalation — so the recording corpus presents a continuous timeline with a silent hole, a §11.4 PASS-bluff at the recording-integrity layer (it claims continuous capture while no data exists for the gap). On a tracked device leaving its reachable state the system MUST, automatically: (a) **DETECT** the drop and **log an honest offline event** into the recording corpus (§11.4.6 — a silent gap presented as continuous capture is a fabricated-continuity bluff; §11.4.128 — a silent recording gap defeats always-on recording); (b) **WAIT** for the device to return using the project's ALREADY-DEFINED reconnection timings — never invented numbers (§11.4.6), the SAME grace / reconnect / poll budgets the project's recovery path already uses; (c) **RE-ATTACH** — resume recording / tracking the moment the device returns and log an honest online / resume event; (d) **ESCALATE** to the project's sanctioned device-recovery path (per §11.4.69 feature class `device_recovery`) if the device does not return within the defined timeout — through the sanctioned, authorization-gated recovery entry point ONLY, never bypassing its gate and never performing a destructive recovery (e.g. a power-cycle) autonomously without that authorization (§11.4.21 / §11.4.101 — high-blast-radius recovery is gated; while blocked the system keeps following the device, logging the blocked-escalation honestly). A tracked-device drop that produces a silent corpus hole, a never-resumed recording, or a never-escalated permanent absence is the bluff this anchor forbids. Composes §11.4.128 (always-on recording — §11.4.144 closes its drop-handling gap) / §11.4.69 (`device_recovery` sink-side positive evidence) / §11.4.6 (honest offline / online events, reused-not-invented timings) / §11.4.14 (watchdog children reaped on stop) / §11.4.21 + §11.4.101 (gated, non-autonomous escalation). Classification: universal (§11.4.17) — the consuming project supplies its concrete tracking transport, device set, already-defined reconnection timings, and sanctioned recovery entry point per §11.4.35. Propagation gate `CM-COVENANT-114-144-PROPAGATION` (literal `11.4.144`) + recommended gate `CM-`

DEVICE-AVAILABILITY-FOLLOWED + paired §1.1 meta-test mutation (strip the watchdog wiring / let an absence go unlogged → gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md §11.4.144. Non-compliance is a release blocker. No escape hatch — no --skip-availability-following, --silent-recording-gap-OK, --no-reconnect-wait, --invent-reconnect-timing, --skip-recovery-escalation, --autonomous-power-cycle-OK flag.

§11.4.145 — Independent multi-angle impact-research per change (User mandate, 2026-06-10). For EVERY fix / change / new feature, INDEPENDENT impact-research agents (subagent-driven §11.4.70/§11.4.20, structurally separate from the author — §11.4.92 self-eval PRECEDES, never satisfies — adversarial "refute-the-change" stance to defeat the documented LLM confirmation-bias failure mode) MUST research the change AND its connected/dependent features across a CLOSED SET OF EIGHT ANGLES — (1) correctness/logic, (2) regression (what existing working feature could break — via call-graph impact enumerating every direct + transitive caller and contract-dependent feature, each mapped to its test), (3) latent/dangerous-code — the recents class (runtime-only failure, interface/ABI/contract mismatch, concurrency/data-race, resource lifecycle), (4) security (taint/injection/secret/unsafe-API), (5) performance (latency/memory/thermal under load vs baseline), (6) host/data/target-hardware safety per §12/§9/§11.4.133, (7) cross-feature interaction (shared state/timing/hardware contention), (8) business-logic conformance (matches the spec AND the connected features' contracts, not merely compiles). Forensic anchor (FACT, project-internal): the recents / 3-button-nav path shipped a latent AIDL-interface-contract mismatch that PASSED code review yet failed at RUNTIME ONLY (ANR on a recents-reaping gesture) because no review angle interrogated the interface contract / concurrency-lifecycle / silently-broken connected feature — the §11.4.108 SOURCE→ARTIFACT→RUNTIME→USER-VISIBLE gap caught only after the fact; §11.4.145 shifts that discovery LEFT. Each angle MUST name the TOOL(S) used + obtain the required DEPTH (the diff, the full changed unit, its declared contract, the spec section, every connected feature — NEVER the diff lines alone) + emit a captured-evidence conclusion per §11.4.5/§11.4.69 ("looks fine" forbidden, §11.4.1); a genuinely-N/A angle is recorded NOT_APPLICABLE: <reason> per §11.4.6, never silently skipped. Output = a per-change impact-research REPORT (one section per angle: tool + evidence path + verdict) + a single GO/NO-GO that BLOCKS acceptance/commit/build on ANY unmitigated risk in ANY angle; the change is fixed/mitigated + affected angles re-researched, iterating to a CLEAN GO (zero unmitigated risk + zero warning) per §11.4.134 before the §11.4.125 review gate. Honest boundary (§11.4.6): a GO proves cross-angle internal-consistency + bounded blast-radius — it does NOT replace §11.4.108 runtime-signature verification nor §11.4.40 full-suite retest; it is one of MULTIPLE STRONG LAYERS, the FIRST research pass every change crosses. Composes/strengthens §11.4.92/.125/.108/.110/.134/.78/.79/.107/.85/.133/§12/§9/.99/.6/§1.1. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-145-PROPAGATION (literal 11.4.145) + recommended gate CM-IMPACT-RESEARCH-PER-CHANGE + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; accept a change with a report missing an angle or an unmitigated NO-GO → CM-IMPACT-RESEARCH-PER-CHANGE FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md §11.4.145. Non-compliance is a release blocker. No escape hatch — no --skip-impact-research, --single-angle-suffices, --author-researches-own-change, --diff-skim-OK, --no-go-overridable, --trivial-change-no-research flag.

§11.4.146 — Reproduce-first test + same-test-confirms-fix + mandatory extend-to-all-cases workflow (User mandate, 2026-06-10). Every reported problem MUST be handled by a NAMED three-step test workflow — §11.4.146 does NOT re-author its component disciplines, it NAMES + BINDS them into the operator's exact per-defect sequence and adds ONLY the two emphases the components leave implicit. **(STEP 1 — REPRODUCE-FIRST + INVESTIGATE)** before any fix, author the §11.4.43 RED test as a §11.4.115 RED-baseline-on-the-broken-artifact (reproduce the defect on the CURRENT pre-fix artifact, capture defect-present physical evidence per §11.4.5/§11.4.69/§11.4.107); NEW EMPHASIS (D1) that same RED test is ALSO a deliberate investigation instrument per §11.4.102(A) — it MUST gather ADDITIONAL forensic data characterising the defect (triggers, boundaries, input/topology scope, adjacent failure modes) that feeds BOTH the fix design AND the STEP-3 extend scope; a RED test used only as a binary present/absent gate with no characterisation satisfies §11.4.115 but NOT §11.4.146 STEP 1. **(STEP 2 — SAME-TEST-CONFIRMS-FIX)** after the fix the SAME test source (the §11.4.115 polarity switch flipped `RED_MODE=1→0`) confirms the defect ABSENT — RED-on-broken then GREEN-on-fixed, both captured, on a clean target per §11.4.108/§11.4.139, validated-first-after-redeploy per §11.4.130, deterministic per §11.4.50; no separate happy-path test substitutes (the §11.4.43/§11.4.115 PASS-bluff). **(STEP 3 — EXTEND-TO-ALL-CASES, mandatory per-fix)** NEW EMPHASIS (D2) immediately after STEP 2 the test MUST be FANNED OUT across the full case-space of the SAME functionality — all flows, valid + invalid + boundary (§11.4.85 empty/max/off-by-one) + concurrent (§11.4.85 contention) + failure-injection (§11.4.85 chaos) + topology variants (§11.4.3) — confirming no issue of any kind, with PROVABLE enumerated coverage per §11.4.118 (a listed case-set with per-case outcome, never "no other issues found"); a REQUIRED step, NOT deferred to a release-cycle discovery sweep and NOT reduced to a single guard; each user-visible case carries rock-solid physical evidence per §11.4.123 + is registered into the §11.4.135 standing regression-guard suite; a newly-discovered fan-out defect triggers §11.4.4 test-interrupt + re-enters STEP 1. (ANTI-BLUFF, all steps) every PASS is rock-solid CAPTURED physical evidence per §11.4.123 (§11.4.5/§11.4.69/§11.4.107) — metadata-only / config-only / absence-of-error / grep-without-runtime PASS forbidden (§11.4/§11.4.1); unclear validation method ⇒ deep-research-before-declaring-untestable per §11.4.123/§11.4.8/§11.4.99; an operator-found defect the green suite missed triggers §11.4.138. Honest boundary (§11.4.6): STEP 3 reduces the unknown-unknown surface but does not prove zero remaining defects (§11.4.118) — un-exercised cases stated as honest gaps, never silently implied clean. Composes §11.4.43/.115/.102/.130/.108/.139/.50/.85/.118/.135/.3/.123/.5/.69/.107/.138/.4/§107/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-146-PROPAGATION` (literal `11.4.146`) + recommended gate `CM-REPRODUCE-FIRST-THEN-EXTEND` (every closed defect's fix carries a §11.4.115 reproduce-first polarity test with captured defect-characterisation [STEP 1] + its `RED_MODE=0` GREEN confirmation [STEP 2] + an enumerated per-functionality extend case-set registered into the §11.4.135 suite [STEP 3]; a closure with the reproduce→confirm pair but NO enumerated extend case-set FAILs) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; close a defect with only the reproduce→confirm pair and no extend-case-set → `CM-REPRODUCE-FIRST-THEN-EXTEND` FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.146. Non-compliance is a release blocker. No escape hatch — no `--skip-reproduce-first`, `--fix-without-red`, `--skip-extend-to-all-cases`, `--reproduce-confirm-suffices`, `--defer-extend-to-release-sweep`, `--single-guard-suffices`, `--no-defect-characterisation` flag.

§11.4.147 — Crashed-agent respawn-until-complete + no-work-loss registry mandate (User mandate, 2026-06-10). Verbatim operator intent: "any agent that crashed because of something MUST BE respawned and finish its work at some point! We MUST NOT lose any work, forget about or have it corrupted!" Forensic case study (FACT): dispatched subagents died on TRANSIENT causes (`API Error: Server is temporarily limiting requests` rate-limit killed 5 at once, `API Error: socket connection closed unexpectedly` killed 2, one left PARTIAL edits — two source files written, the dependent SystemUI sliders + doc + test NOT done), each re-dispatched by pure conductor vigilance with NO mechanical guarantee — the moment conductor context is lost / compacted / the conductor itself crashes, an in-flight-but-dead work unit is silently dropped (lost / forgotten / corrupted). Every dispatched agent/subagent MUST be tracked through its full lifecycle so a crash NEVER loses, forgets, or corrupts its work; a crashed agent is NOT a completed agent (§11.4.6 — "it probably finished before dying" is a forbidden guess; abnormal termination is positive evidence the unit is OPEN). The mandate (ALL hold): **(a) durable agent REGISTRY** — every dispatched agent has an append-only machine-readable registry entry (stable id, task + §11.4.58 file-scope, declared output path(s) + §11.4.5/§11.4.69 evidence dir, dispatch timestamp, status from the CLOSED SET `{dispatched | in-flight | crashed | respawned | complete}`), reusing the §11.4.116 sync substrate (JSONL event stream + atomic status snapshot; "an entry with no dispatch-event cannot show `complete`"); the registry is the SINGLE SOURCE OF TRUTH for "is this work owed?" and an unregistered dispatch is itself a violation; **(b) mechanical CRASH-DETECTION + RESPAWN-until-complete** — any abnormal termination (transient rate-limit/socket-close, any non-completion exit, or a terminal error after the runtime's own retries are exhausted) flips the entry to `crashed` + keeps the unit OPEN, and the unit is respawned (fresh agent claims the same id + scope + preserved state) and re-respawned until an agent reaches `complete`; respawn is the safe/reversible/bounded decision taken autonomously per §11.4.101 (never blocks the loop for a human), transient causes use the project's ALREADY-DEFINED backoff budgets (never invented, §11.4.6), a deterministically-reproducible non-transient terminal error is investigated per §11.4.102 before further respawn (never a blind retry loop); **(c) PARTIAL-STATE preserve → §11.4.84-check → resume-or-clean-restart** — the crashed agent's uncommitted edits + output doc are PRESERVED (never silently discarded → lost, never blindly committed → corruption), the respawn runs the §11.4.84 quiescence check on the preserved tree (every modified file accounted-for vs scope, no mutation/`// always pass`/`_mutated_*` residue, no half-written/torn artifact), then EITHER RESUMES idempotently when it passes OR CLEAN-RESTARTS from a known-good base (§9.2 pre-op backup, reversible §11.4.101) when inconsistent — nothing lost, nothing corrupted; per-agent `git worktree` isolation (§11.4.58 L4 / §11.4.84) keeps the partial tree from contaminating other streams; **(d) "crash ≠ done" COMPLETION criterion** — a unit is DONE only when an agent reaches `complete` with its required captured evidence/output landed (§11.4.5/§11.4.69 + the §11.4.116 verdict-carries-evidence-path rule); the endless-loop done-condition (§11.4.87/§11.4.94/§11.4.97/§11.4.126) MUST NOT read satisfied while any entry is `dispatched`/`in-flight`/`crashed`/`respawned`-not-yet-`complete`, the zero-idle survey (§11.4.94) treats every non-`complete` entry as an OPEN item to reclaim, and a registry showing `complete` without landed evidence is a §11.4 PASS-bluff at the agent-lifecycle layer. Honest boundary (§11.4.6): the registry + respawn guarantee work is not lost/corrupted, NOT that it is correct — the respawned output still crosses §11.4.108 + §11.4.125/§11.4.142 review + §11.4.40 retest; §11.4.147 is the durability layer beneath those, the agent-side analogue of §11.4.144 (same detect→wait/backoff→re-attach/respawn→escalate shape). Composes

§11.4.6/.58/.84/.87/.94/.97/.101/.116/.102/.108/.125/.142/.40/.128/.144/§9.2/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-147-PROPAGATION` (literal `11.4.147`) + recommended gate `CM-CRASHED-AGENT-RESPAWN-TRACKED` + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; mark a `crashed` entry as the loop's done-condition `complete` without landed evidence, OR drop a dispatched agent without a registry entry → `CM-CRASHED-AGENT-RESPAWN-TRACKED` FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.147. Non-compliance is a release blocker. No escape hatch — no `--skip-agent-registry`, `--crash-equals-done`, `--no-respawn`, `--discard-partial-state`, `--blind-commit-partial`, `--forget-dead-agent`, `--loop-done-with-crashed-entries` flag.

§11.4.148 — Workable-item integrity (status+type+id) + comprehensive structured description + bidirectional external-tracker sync + BLOCKED unblock-choices mandate (User mandate, 2026-06-10). BINDS + STRENGTHENS the workable-item discipline (§11.4.15 status / §11.4.16 type / §11.4.54 id / §11.4.21 Operator-blocked / §11.4.91 description-clarity / §11.4.93 + §11.4.95 SQLite SSoT / §11.4.106 docs_chain) into ONE integrity contract spanning DB ? docs ? external tracker, adding the operator's three emphases: **(D1) no item without a valid status + valid type + stable id — on ALL three surfaces** (an item missing any of the three in the DB, the rendered docs, OR the external tracker FAILs the validator — release-blocker; the id is the cross-surface binding key); **(D2) comprehensive structured description per item** — WHAT it is (§11.4.91 ≥6-word/≥40-char clear meaning) + HOW it manifests + HOW to reproduce (§11.4.115/.146) + ACCEPTANCE CRITERIA (§11.4.123/.69 captured-evidence verdict), in the §11.4.93 DB `description` column AND the docs AND the tracker, never a stub/§11.4.91-anti-pattern fragment; **(D3) BLOCKED items carry WHY + enumerated unblock CHOICES** — tightens §11.4.21: every `Operator-blocked` item (incl. `Blocked`/`BLOCKED` documented alias normalised to canonical `Operator-blocked`, never a silent fork §11.4.6) MUST enumerate the closed list of decisions/actions that would unblock it (`[A]…· [B]…· [C]…`, mirroring §11.4.66's 2–4-option shape); a BLOCKED item with no enumerated choices FAILs; **(D4) regular never-missed bidirectional DB?docs?tracker sync** — the §11.4.93/.95 git-tracked SQLite DB is the SSoT, docs + tracker DERIVED; full sync (`db-to-md` / `md-to-db` byte-identical round-trip §11.4.93 + tracker push) runs regularly + on every change, §11.4.86 drift-proof fingerprint (sha256 of the sorted item keyset, NOT mtime) gates freshness, §11.4.106 docs_chain-bound so drift is mechanically caught not vigilance-dependent; **(D5) generic external-tracker push** carries statuses (collapsed onto the tracker's native set per §11.4.33/.112 when fixed, precise value preserved in a header, never lost), types, assignee (§11.4.104 participant handle, UNSET defaults from a project env var, never hardcoded/logged §11.4.10), and sub-tasks, IDEMPOTENT (dry-run-then-real, match-by-stable-key `[<ID>]` prefix / id custom-field, present⇒UPDATE/absent⇒CREATE, rate-limited, credential-redacted, sink-side `created=N updated=M failed=0` proof §11.4.69), the MACHINERY project-agnostic §11.4.28 (consumer registers its tracker/list/field-map at runtime). Anti-bluff §11.4: every sync + validator pass carries captured evidence §11.4.5/.69; honest boundary §11.4.6 — guarantees well-formed items + agreeing surfaces, NOT that the underlying work is correct (still crosses §11.4.108/.40/.123). Composes §11.4.15/.16/.21/.33/.34/.54/.66/.86/.91/.93/.95/.104/.106/.112/.123/.10/.28/.69/.6/§1.1. Classification: universal (§11.4.17) — consumer supplies its DB path, id prefix, tracker service + list/board id + field map + default-assignee env var, docs_chain context per §11.4.35. Propagation gate `CM-COVENANT-114-148-`

PROPAGATION (literal 11.4.148) + recommended gates CM-ITEM-INTEGRITY-STATUS-TYPE-ID / CM-ITEM-COMPREHENSIVE-DESCRIPTION / CM-BLOCKED-UNBLOCK-CHOICES / CM-TRACKER-SYNC-IDEMPOTENT + paired §1.1 mutation (gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md §11.4.148. Non-compliance is a release blocker. No escape hatch — no --item-without-status, --item-without-type, --item-without-id, --stub-description-OK, --blocked-without-choices, --skip-tracker-sync, --one-way-sync-OK, --non-idempotent-tracker-push, --hardcode-assignee flag.

§11.4.149 — Per-workable-item testing-diary mandate (User mandate, 2026-06-10). Every workable item MUST carry an append-only TESTING DIARY — a chronological record of every test run against it — distinct from §11.4.93 item_history (lifecycle STATE transitions) + §11.4.55 Reopens (reopen cycles): the diary records TEST EXECUTIONS (which may or may not change status). The mandate (ALL hold): **(a)** a test_diary table additive to the §11.4.93/.95 SQLite SSoT, one row per run soft-keyed to the item id, capturing ISO-8601-UTC date_time + tested_by (closed set User|Operator|AI-agent|HelixQA) + result (§11.4.45 PASS|FAIL|SKIP + detail) + in-depth observations (long markdown, facts or explicit UNCONFIRMED:/PENDING_FORENSICS: §11.4.6, the background a future fix needs) + action_taken+status_changed+from/to (did this run change status + WHY) + §11.4.69 evidence_path+feature_class, with a SCHEMA CONSTRAINT making a PASS row WITHOUT a non-empty evidence path impossible (a PASS-bluff rejected by the schema itself); **(b)** BOTH an in-depth diary doc + a derived at-a-glance summary VIEW (total/pass/fail/skip + last-verdict + last-run + status-changes + distinct testers/feature-classes, DERIVED never duplicated §11.4.93) feeding §11.4.132 risk-ordering; **(c)** four-format per-item Diary/Diary_Summary exports §11.4.65 + §11.4.86-drift-proof-fingerprinted (sha256 of the sorted diary keyset) + §11.4.106 docs_chain-bound so a diary row not exported/pushed FAILs a gate (never-missed); **(d)** external-tracker SUB-TASK model — the item task's description stays the item (§11.4.148 D2, NOT polluted with diary text), each run a CHILD sub-task (type task) with a {TODO|In-progress|Completed} lifecycle (collapsed per §11.4.33/.112), observations + action + an Evidence: line (path not raw artefact §11.4.10/.13) + a diary-entry idempotency key, reusing the §11.4.148 D5 idempotent rate-limited credential-redacted sink-side-proven push, mapper project-agnostic §11.4.28; **(e)** MINIMAL-LLM deterministic bash/Go tooling (zero LLM in the data path — the observations prose is authored by whoever ran the test; tooling only stores/renders/pushes/validates), 100% test-covered §11.4.27 (unit + integration-against-the-real-tracker with an honest §11.4.3 SKIP-when-token-absent never a faked PASS + export + HelixQA Challenge + paired §1.1 mutation) so a diary PASS-bluff is mechanically impossible. Honest boundary §11.4.6 — the diary guarantees a complete auditable per-item test-history, NOT that any single run's verdict is correct (rests on its own §11.4.69/.107/.123 evidence). Composes §11.4.6/.27/.45/.50/.55/.65/.69/.86/.93/.95/.106/.107/.123/.132/.148/.10/.13/.28/.33/.112/§1.1. Classification: universal (§11.4.17) — consumer supplies its DB path, diary doc layout, export formats, tracker sub-task field map, docs_chain context per §11.4.35. Propagation gate CM-COVENANT-114-149-PROPAGATION (literal 11.4.149) + recommended gates CM-TEST-DIARY-SYNC / CM-DIARY-PASS-REQUIRES-EVIDENCE + paired §1.1 mutation (gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md §11.4.149. Non-compliance is a release blocker. No escape hatch — no --skip-testing-diary,

`--diary-without-evidence`, `--no-diary-summary`, `--diary-without-tracker-subtask`, `--llm-in-diary-data-path`, `--diary-export-optional` flag.

§11.4.150 — Mandatory deep multi-angle web research per change/issue, before declaring fixed or structural (User mandate, 2026-06-11). Verbatim operator mandate: "For every single issue we fix or improvement we make — besides tight systematic-debugging, fixing, code review by independent agents, comprehensive tests — we MUST ALWAYS do deep web research from various angles! No matter how big/small/simple/complex, dig deep the internet for articles, technical documentation, APIs, open-source code — EVERYTHING that can help make the best possible solution OR confirm we don't have a more serious problem we're unaware of! We MUST do everything possible to STOP the constant issue-reopening and finally start closing items as fixed+working! Do this ALWAYS in parallel with the main work stream!" For EVERY fix / improvement / change / closure — no matter how big, small, simple, or complex — the agent MUST, IN ADDITION to tight systematic-debugging (§11.4.102), the fix, independent-agent code review (§11.4.125 / §11.4.142), and comprehensive multi-layer tests (§11.4.4(b) / §11.4.40), perform a DOCUMENTED **deep multi-angle web research pass** digging the internet from VARIOUS angles — articles, official + vendor technical documentation, API references, standards, issue trackers, maintainer guidance, reusable open-source code — to BOTH (i) discover the best possible solution AND (ii) confirm there is NOT a more serious underlying problem the team is unaware of. Forensic case study (FACT, 2026-06-11): a subtitle-on-secondary-display goal verdicted `Won't-fix: structurally-impossible` (§11.4.112 secure-surface pixel-blanking) was OVERTURNED by a deep multi-angle research pass that surfaced the real OCR-of-the-PRIMARY-display path — a "we can't" became a shipped capability; the canonical anti-pattern of a structural verdict reached for want of research. The mandate (ALL hold): **(A) No closure-as-fixed/structural WITHOUT a documented deep-research pass** — no item may be marked `Fixed`/`Implemented`/`Completed` (§11.4.33) OR classified `structurally-impossible` won't-fix (§11.4.112) until a deep multi-angle pass is documented; §11.4.150 makes §11.4.8's pass UNCONDITIONAL (every item, however trivial, at the closure AND structural-verdict gates). **(B) Multiple angles, not a single lookup** — ≥ 2 genuinely-distinct angles (official-docs / standards / known-bug-trackers / alternative-approach / failure-mode / security / performance / platform-constraint / open-source-precedent) so a single-source confirmation-bias miss (§11.4.145) cannot pass; a one-link drive-by is NOT deep multi-angle. **(C) Confirm-no-bigger-problem, not just find-a-fix** — explicitly seek evidence the fix does not mask a deeper defect AND the closure/verdict is genuinely safe; "we found nothing worse" requires the enumerated search (§11.4.118). **(D) Latest-source + cited** — LATEST authoritative versions per §11.4.99 (never training-data/memory), each cited by URL + access date in the research artefact AND the closure commit footer (`Deep-research <date>: <urls>` OR the literal `NO external solution found — original work` per §11.4.8). **(E) Reopen-breaking is the PURPOSE** — STOP the constant Fixed→Reopened churn (§11.4.34 / §11.4.55); a reopen whose root cause a deep pass would have surfaced is a §11.4.150 miss; composes §11.4.7 (demotion-evidence) + §11.4.112 (structural verdict now REQUIRES the cited-authorities pass). **(F) ALWAYS in parallel with the main stream** — background subagent-driven (§11.4.70 / §11.4.20 / §11.4.103 / §11.4.89) concurrent with main fix/build/test work, NEVER serialising it or stalling the loop (§11.4.94 / §11.4.97 / §11.4.101 / §11.4.126). **(G) Apply ASAP to every workable item** — retroactively + going forward across the §11.4.93 / §11.4.95 SSoT; items closed without a documented pass are re-audited in the §11.4.40 / §11.4.42 release-gate sweep.

Honest boundary (§11.4.6): the pass reduces the unknown-unknown surface + breaks the most common reopen causes — it does NOT prove zero remaining defects (§11.4.118) and does NOT replace §11.4.108 runtime-signature verification, §11.4.125 / §11.4.142 review, or §11.4.40 retest; it is the research layer every fix/closure/structural-verdict additionally crosses, and "we probably don't have a bigger problem" without the enumerated multi-angle search is a guess (§11.4.6), never a finding. Composes §11.4.8 / §11.4.99 / §11.4.123 / §11.4.118 / §11.4.145 / §11.4.125 / §11.4.142 / §11.4.7 / §11.4.112 / §11.4.34 / §11.4.55 / §11.4.70 / §11.4.20 / §11.4.89 / §11.4.103 / §11.4.40 / §11.4.42 / §11.4.93 / §11.4.95 / §11.4.6 / §1.1. Classification: universal (§11.4.17) — the consuming project supplies its concrete research corpora, angle set, item tracker, and closure-commit-footer convention per §11.4.35. Propagation gate `CM-COVENANT-114-150-PROPAGATION` (literal `11.4.150`) + recommended gate `CM-DEEP-RESEARCH-PER-ISSUE` (every closed/structural-verdicted item carries a documented multi-angle deep-research artefact + cited-source closure footer, run in parallel, before the closure/structural verdict is accepted) + paired §1.1 mutation (strip the literal → propagation gate FAILs; close an item or reach a `structural` verdict with no documented multi-angle research artefact / cited footer → `CM-DEEP-RESEARCH-PER-ISSUE` FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.150. Non-compliance is a release blocker. No escape hatch — no `--skip-deep-research`, `--single-source-suffices`, `--trivial-change-no-research`, `--close-without-research`, `--structural-without-research`, `--serialise-research`, `--research-from-memory-OK` flag.

§11.4.151 — Project-prefixed release-tag/version-naming mandate (User mandate, 2026-06-12). Verbatim operator mandate: "Every release tag and version name we create — on the main repository and on every Submodule we own — MUST be prefixed with the project's release prefix, e.g. `helix_translate-1.0.0-dev-0.0.1`. The prefix MUST come from `HELIX_RELEASE_PREFIX` in our `.env` if it is set, otherwise from the lowercased project root directory name. The SAME prefix MUST be used across the main repo and all owned Submodules in one release so a release is greppable across every repository." Every release tag AND every version name created on the main repository AND on every owned-by-us submodule (§11.4.28) MUST be prefixed with the project's release prefix, form `<PREFIX>-<version>` (canonical example: `helix_translate-1.0.0-dev-0.0.1`), so a release is identifiable + greppable across every repository it spans (one `git tag -l '<PREFIX>-*'` enumerates the whole release surface). **Prefix resolution order (closed-set, deterministic — §11.4.6 no-guessing):** (1) `HELIX_RELEASE_PREFIX` from the project's `.env` — authoritative when set; `.env` is git-ignored per §11.4.30 and the variable is documented in the tracked `.env.example` (a §11.4.77 re-obtain mechanism, never committed); (2) fallback = the lowercased snake_case form of the project root directory name (no spaces) per §11.4.29 — used whenever the env var is unset/empty, so a prefix is ALWAYS resolvable from the checkout with zero operator input. **The prefix MUST be IDENTICAL across the main repo and all owned submodules within a single release** — a release that tags the main repo `<PREFIX>-1.2.0` while tagging an owned submodule with a different/unprefixed value is a §11.4.151 violation (the cross-repo grep no longer enumerates the release). Version codes increment monotonically within the prefix (`<PREFIX>-…-0.0.1` → `<PREFIX>-…-0.0.2` → …), never reset, never skipped. Honest boundary (§11.4.6): the prefix guarantees a release is identifiable + uniform across every repository, NOT that its contents are correct — the tag is still created only after the §11.4.40 full-suite retest GREEN and reaches every upstream via the §11.4.113 merge-onto-latest-main path (NEVER a force-push), fanned out per §2.1. Composes §2.1 / §11.4.29 / §11.4.30 / §11.4.40 /

§11.4.113 / §11.4.126 (the release-scope terminal condition is a published, prefixed tag). Classification: universal (§11.4.17) — the consuming project supplies its concrete prefix value + the `HELIX_RELEASE_PREFIX` env var per §11.4.35. Propagation gate `CM-COVENANT-114-151-PROPAGATION` (literal `11.4.151`) + recommended gate `CM-RELEASE-PREFIX-NAMING` (every release tag/version on the main repo + every owned submodule carries the resolved `<PREFIX>-` prefix, identical across the release) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; create an unprefixed or differing-prefix release tag → `CM-RELEASE-PREFIX-NAMING` FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.151. Non-compliance is a release blocker. No escape hatch — no `--no-release-prefix`, `--unprefixed-tag`, `--prefix-optional`, `--differing-submodule-prefix` flag.

§11.4.152 — Crashlytics-recorded-data continuous monitoring + systematic-debug + regression-test-coverage mandate (User mandate, 2026-06-13). Verbatim operator intent: "For every project that has Firebase Crashlytics enabled / wired, we MUST continuously monitor ALL of the Crashlytics-recorded data — crashes, ANRs, performance traces, and non-fatals — systematically debug each, fix and improve, and cover everything with validation and verification tests. This MUST be checked regularly, with no false results and no bluff of any kind!" Every project that has Firebase Crashlytics enabled/wired (SDK linked into a shipping artifact, crash+non-fatal+ANR reporting active) MUST treat the Crashlytics console as a first-class captured-evidence channel from real end-user devices and continuously process every datum it records — an open Crashlytics issue with a green test suite is a §11.4 PASS-bluff at the field-telemetry layer (a real user hitting a broken feature while the suite reports green). STRENGTHENS+COMPLETES §11.4.47: §11.4.47 owns the periodic REVIEW + dedup + Issue-creation pass that SURFACES Crashlytics/Analytics/Performance findings; §11.4.152 owns what happens to each surfaced item AFTER — the systematic-debug → fix/improve → regression-test-coverage lifecycle that drives it to a proven regression-immune closure (§11.4.47 finds it; §11.4.152 fixes it + proves it stays fixed; both mandatory, neither substitutes). The mandate (ALL hold): (1) **continuous monitoring of ALL four surfaces** — fatal crashes, ANRs, performance traces/regressions, AND non-fatals (skipping any one is a PASS-bluff; non-fatals are the silent class — every catch/fallback/recovered-error path is a real degraded UX and MUST be triaged, not ignored because the app did not crash); (2) **systematic-debugging of each (reproduce-before-fix)** per §11.4.102 Iron Law + §11.4.115 RED-baseline-on-the-broken-artifact (the recorded stacktrace/ANR-trace/non-fatal-context IS the §11.4.5/.69 captured defect-present evidence) BEFORE the fix — a fix without a prior reproducing test is a §11.4.43/.123 violation; unreproducible ⇒ deep-research-before-declaring-untestable §11.4.123/.150; (3) **a fix/improvement for every confirmed issue** against its proven root cause (§11.4.9/.43/.108), "acknowledged in console" is not a fix, muting a recurring issue without a tracked rationale is a §11.4.6/.90 violation; (4) **validation+verification regression-test coverage per closed issue** — in the SAME commit as the fix register a permanent §11.4.135 regression guard (§11.4.115 polarity test: `RED_MODE=1` captures the recorded defect on the pre-fix artifact, `RED_MODE=0` standing GREEN guard asserting ABSENT) exercising the same user-reachable path the stacktrace identifies, with rock-solid captured evidence §11.4.5/.69/.107/.123 + a closure log citing the console issue id/URL + root-cause + fix commit + the validation/verification test paths; a Crashlytics issue marked resolved WITHOUT a falsifiable regression test is FORBIDDEN (the silent-recurrence vector, §11.4.138 operator-escape class applied to field telemetry); (5) **regular cadence** reusing the §11.4.47 five-trigger set (pre-build/pre-flash/pre-distribute/pre-tag blocking; daily + post-deployment-burn-in non-blocking) — a

one-time sweep never repeated is a violation; (6) **no false results, no bluff** (§11.4/.1/.6 — "no new issues" requires the enumerated monitored-surfaces+window §11.4.118; "fixed" requires the RED→GREEN flip §11.4.115/.130; a guard whose §1.1 mutation does not FAIL it is a bluff gate). Honest boundary §11.4.6: processing every recorded issue breaks the silent-recurrence vector — it does NOT prove zero remaining field defects nor replace §11.4.108/.40; an issue is "closed" only when its guard is GREEN on a clean artifact, never when the console mark is flipped. Classification: universal (§11.4.17) — the consuming project supplies its Crashlytics handle, console-access credential (§11.4.10, never logged), severity table, and per-issue closure-log path per §11.4.35; the reference project-level instantiation is the Lava client's §6.O (Crashlytics-Resolved Issue Coverage — per-issue validation test + Challenge test + `.lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md` closure log) + §6.AC (Comprehensive Non-Fatal Telemetry — every catch/fallback records a non-fatal with triage context). Composes §11.4/.1/.5/.6/.9/.34/.40/.43/.47/.69/.90/.102/.107/.108/.115/.118/.123/.130/.135/.138/.150/§1.1. Propagation gate `CM-COVENANT-114-152-PROPAGATION` (literal `11.4.152`) + recommended gate `CM-CRASHLYTICS-ISSUE-FULLY-COVERED` (every closed Crashlytics issue carries a systematic-debug root-cause record + a registered §11.4.135 regression guard + a closure-log entry citing the console issue id/URL + the validation/verification test paths; a console-resolved issue with no falsifiable regression guard FAILs) + paired §1.1 meta-test mutation (gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.152. Non-compliance is a release blocker. No escape hatch — no `--skip-crashlytics-monitoring`, `--monitor-crashes-only`, `--skip-non-fatals`, `--resolve-without-regression-test`, `--console-mark-is-fixed`, `--monitor-once`, `--mute-without-rationale` flag.

§11.4.153 — Comprehensive per-feature Status + Status_Summary document set with mandatory video-recording confirmation (User mandate, 2026-06-15). Every project MUST maintain, under `docs/features/`, a comprehensive feature Status document set (`Status.md` + its §11.4.56 `Status_Summary.md` companion) enumerating EVERY system component, EVERY client app/binary/surface (TUI / CLI / Web / desktop / mobile / API / gRPC / library / submodule / infrastructure), and EVERY feature — including features ported from any incorporated CLI-agent / submodule catalogue (§11.4.74) — with NO single feature left out. (1) **Total categorized coverage** — per-feature table organized Component→Category, reconciled against the actual codebase (a code-present feature missing from the table, or a row with no code, is a §11.4.6/§11.4.118 bluff). (2) **Per-feature fields** — Component / Feature / Category / Implementation / Wiring (genuinely end-user-reachable, not merely compiled, §11.4.108) / Real-use / Tests-coverage (four-layer §11.4.4(b)) / Validation (PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED per §11.4.45) / **Video-recording confirmation** (path to the real-use video or honest gap marker). (3) **Mandatory per-feature real-use video** — every user-visible "confirmed" claim backed by a recorded real-use video of a genuine end-user scenario (real prompts → real LLM/service responses → real results), NEVER a frozen/stale frame (§11.4.107), NEVER faked/mocked/demo-loop/bluff response or LLM error passed off as success (§11.4.2/§11.4.5), stored at the project-declared recording path (§11.4.35); a confirmed row with no real video or a bluff video = §11.4 PASS-bluff; autonomous-infeasible ⇒ honest §11.4.3/§11.4.52 SKIP + tracked migration item, NEVER a faked confirmation. (4) **Video-analysis remediation loop** — every video analysed; any defect it surfaces triggers §11.4.102 systematic-debug → fix → §11.4.146 retest → re-record → clean GO per §11.4.134 before "confirmed" (a video exposing a broken feature is a §11.4.4 test-

interrupt). (5) **Always-in-sync** — §11.4.45-class roster/corpus-backed, §11.4.106 docs_chain-bound + §11.4.86 drift-proof fingerprint (sha256 of sorted feature-key roster AND sorted video-artefact roster, NOT mtime), re-syncs out-of-the-box; stale = violation. (6) **Four-format export** — HTML + PDF + **DOCX** (this doc class ADDS DOCX to the §11.4.65 HTML+PDF set; other classes unchanged), in sync per §11.4.60. (7) Follows §11.4.44/.45/.56/.57/.59/.60. Honest boundary §11.4.6 — guarantees a complete video-confirmed always-synced ledger, NOT per-feature correctness (rests on each row's §11.4.69/.107/.123 evidence) and NOT a §11.4.40 retest substitute. Classification: universal (§11.4.17). Composes §11.4.2/.5/.44/.45/.52/.56/.57/.59/.60/.65/.86/.102/.106/.107/.108/.118/.123/.134/.146/§1.1. Propagation gate `CM-COVENANT-114-153-PROPAGATION` (literal `11.4.153`) + recommended gates `CM-FEATURE-STATUS-COMPLETE` + `CM-FEATURE-STATUS-VIDEO-CONFIRMED` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.153. Non-compliance is a release blocker. No escape hatch — no `--skip-feature-status`, `--feature-without-video`, `--frozen-video-OK`, `--bluff-response-video-OK`, `--skip-video-analysis`, `--feature-ledger-incomplete-OK`, `--no-docx-export`, `--allow-feature-ledger-drift` flag.

§11.4.154 — Window-scoped capture + fresh-corpus rotation for feature/QA recordings (User mandate, 2026-06-15). Verbatim: "recording you perform does record the window containing apps and services, not the whole desktop or monitor screen!" + "All old recording files MUST BE removed when new one starts!" Refines §11.4.2/.5/.107/.153 recording discipline with two capture-hygiene invariants. **(A) Window-scoped, NOT whole-screen** — every feature/QA video MUST capture ONLY the window/surface of the app/service under test (GUI window / CLI-TUI terminal pane / web tab-viewport / device-emulator-simulator frame), NEVER the whole desktop/monitor or unrelated windows; whole-desktop capture leaks operator-private content (§11.4.10/.83), dilutes the §11.4.107 liveness/freeze oracle, and breaks the §11.4.137 OCR/ROI content oracle. Target the window/region by stable identity (window id/title, device serial, browser context, tmux target) per §11.4.111 — never a fixed full-screen device index. Platform genuinely cannot capture below whole-screen ⇒ honest §11.4.3 SKIP + tracked migration item, never a whole-screen pass-off. **(B) Fresh-corpus rotation** — when a new recording run for a scope begins, the agent's OWN prior in-scope stale recordings at the raw recording path MUST be removed FIRST so the live corpus reflects the current run (§11.4.107 not-stale + §11.4.86 roster-freshness). Honest boundary (§11.4.6 + §9.2): "remove old" = the agent's own prior recordings for the SAME scope/project ONLY — NEVER another project's/scope's/operator-authored files; uncertain ⇒ surface, don't delete (§11.4.122); committed `docs/qa/<run-id>/` evidence (§11.4.83) is the durable record, NOT rotated. Classification: universal (§11.4.17). Composes §11.4.2/.5/.10/.83/.86/.107/.111/.122/.128/.137/.153/§9.2/.6. Propagation gate `CM-COVENANT-114-154-PROPAGATION` (literal `11.4.154`) + recommended gate `CM-WINDOW-SCOPED-FRESH-CORPUS-RECORDING` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.154. Non-compliance is a release blocker. No escape hatch — no `--whole-screen-capture-OK`, `--skip-window-scope`, `--keep-stale-recordings`, `--no-corpus-rotation`, `--full-desktop-recording` flag.

§11.4.155 — Project-name-prefixed feature/QA recording filenames (User mandate, 2026-06-15). Verbatim: "All recorded videos MUST START with prefix: the PROJECT NAME (ALWAYS USE THE PROJECT NAME). Project name

MUST be obtained according to the constitution's own project-name resolution." Every recorded video the project produces — every feature/QA real-use recording (§11.4.153), every window-scoped capture (§11.4.154), every always-on device recording (§11.4.128), every raw or curated artefact at the project-declared recording path (§11.4.35) + the committed `docs/qa/<run-id>/` trail (§11.4.83) — MUST have a filename that STARTS WITH the PROJECT-NAME prefix, ALWAYS; an unprefixed recording is a §11.4.155 violation (a multi-project/scope corpus on one host per §11.4.128/.103 becomes un-greppable + un-attributable — the §11.4.151 identify-and-grep failure on the recording-corpus axis). **Prefix resolution (closed-set, deterministic — §11.4.6, IDENTICAL to §11.4.151):** (1) `HELIX_RELEASE_PREFIX` from `.env` (authoritative, git-ignored §11.4.30, documented in tracked `.env.example` §11.4.77) else (2) lowercased snake_case project-root dir name §11.4.29 — ALWAYS resolvable, zero operator input. SAME prefix for EVERY recording in a checkout so `ls '<PREFIX>---'*` enumerates the corpus; canonical form `<PREFIX>---<feature-or-scope>---<run-id>.<ext>` (`---` delimits the prefix unambiguously); MUST equal the §11.4.151-resolved release-tag prefix (divergence is itself a §11.4.155 violation — one project, one name). Honest boundary (§11.4.6): the prefix guarantees attribution + greppability, NOT content validity (still §11.4.107/.137/.153) and does NOT relax §11.4.154's window-scope/rotation (rotation removes the agent's OWN `<PREFIX>---*` only; foreign/operator files surfaced not deleted §11.4.122/§9.2). Classification: universal (§11.4.17). Composes §11.4.151/.128/.153/.154/.111/.83/.6/.29/.30/.35/.77/.86/§1.1. Propagation gate `CM-COVENANT-114-155-PROPAGATION` (literal `11.4.155`) + recommended gate `CM-RECORDING-PROJECT-NAME-PREFIX` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.155. Non-compliance is a release blocker. No escape hatch — no `--no-recording-prefix`, `--recording-without-project-name`, `--unprefixed-recording`, `--prefix-optional-for-recording`, `--differing-recording-prefix` flag.

§11.4.156 — All CI/CD automation (GitHub Actions / GitLab pipelines / equivalents) MUST be disabled (User mandate, 2026-06-15). Verbatim operator mandate: "Any GitHub actions or GitLab pipelines MUST BE disabled! Add this critical mandatory rule / mandatory constraint into the root constitution Submodule, commit and fetch all its changes to all upstreams and make sure we respect and follow this rule (we do apply it) ASAP!!!" Every repository this Constitution governs — main repo, this constitution submodule, every owned + nested submodule we author and push — MUST ship with ALL server-side CI/CD automation DISABLED: no push to any owned upstream may trigger a GitHub Actions run, GitLab pipeline, or equivalent (Jenkins/CircleCI/Travis/Drone/Woodpecker/Bitbucket/Azure, any `on: push`/`schedule`/`workflow_dispatch`). GENERALISES + makes ABSOLUTE the §11.4.75 Layer-5 posture (remote CI DISABLED, workflow preserved at a `…disabled-local-only` non-`.yml` name a provider ignores) across ALL governed repos; enforcement migrates to the LOCAL §11.4.75 git-hook ritual + §11.4.40 pre-tag sweep, never a remote runner. ALL hold: **(A)** zero active `.github/workflows/*.yml|yaml` / `.gitlab-ci.yml` / `.gitlab/**` / equivalent at the ROOT of any governed repo/submodule (the only place a provider executes); **(B)** "disabled" = a push triggers ZERO runs — delete OR rename to a non-trigger name (§11.4.75 `.disabled/`.`disabled-local-only`); a live-`on:`+`if:false` workflow still queues runs, NOT compliant; **(C)** scope = repos we author+push — vendored/third-party nested configs below the root (AOSP `external/**`, `prebuilts/**`, vendored submodules) are INERT (a provider never runs a non-root config), OUT of scope, MUST NOT be mass-edited (§11.4.29 vendor-exempt); test = "does a push to OUR upstream trigger a run?" yes⇒disable, inert⇒document+leave (§11.4.6 verify-not-assume); **(D)** no new CI may be added

(release blocker); **(E)** pre-push verify `git ls-files | grep -E '^\.github/workflows/.*\.ya?ml$|^\.gitlab-ci\.yml$'` empty for authored repos, §11.4.109-class PreToolUse guard + gate enforce mechanically. Honest boundary (§11.4.6): file-level disabling stops FILE-triggered runs, NOT provider-side server settings (org-default required workflows, branch-protection required checks, provider scheduled exports) — the operator turns those off; the agent documents what it cannot reach, never claims unachieved completeness. Composes §11.4.75/.29/.6/.40/.42/.109/.113/§2.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-156-PROPAGATION` (literal `11.4.156`) + recommended gate `CM-NO-ACTIVE-CI` + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; add a root `.github/workflows/x.yml` to an authored repo → `CM-NO-ACTIVE-CI` FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.156. Non-compliance is a release blocker. No escape hatch — no `--allow-ci`, `--enable-workflow`, `--keep-pipeline`, `--remote-ci-OK`, `--ci-exempt` flag.

§11.4.157 — GEMINI.md maintained in lockstep with CLAUDE.md / AGENTS.md / QWEN.md (User mandate, 2026-06-15). Verbatim operator mandate: "Make sure with CLAUDE.md, AGENTS.md, QWEN.md we maintain GEMINI.md too! Add this mandatory fact / rule to root constitution Submodule we are inheriting / extending - CONSITUTION.md, CLAUDE.md, QWEN.md, AGENTS.md, GEMINI.md and other related relevant files!" Forensic FACT (2026-06-15): when §11.4.156/§11.4.157 were authored, Constitution/CLAUDE/AGENTS/QWEN carried the family through §11.4.155 but GEMINI.md had silently drifted to §11.4.141 — 14 mandates (§11.4.142–155) never propagated — a §11.4 propagation-bluff (a Gemini-CLI agent reads a stale Constitution). GEMINI.md is a FIRST-CLASS governance context carrier EQUAL to CLAUDE.md/AGENTS.md/QWEN.md, never optional/best-effort. ALL hold: **(A)** five-carrier lockstep — no governance change is complete until GEMINI.md carries it alongside the other three mirrors; GEMINI.md is added to the §11.4.26 propagation + cross-reference set explicitly; **(B)** no silent drift — GEMINI.md lagging the other mirrors' highest rule is a §11.4.157 violation (§11.4.65-class), back-fill required; **(C)** equal status — GEMINI.md restates the SAME literal `11.4.N` anchors the propagation gates require (§11.4.35), fleet count INCLUDES GEMINI.md; **(D)** consumer projects' own CLAUDE/AGENTS/QWEN/GEMINI bind too (§11.4.35). Honest boundary (§11.4.6): the §11.4.142–155 GEMINI.md back-fill is a tracked release-blocking remediation; claiming GEMINI.md "in sync" while the back-fill is incomplete is itself a §11.4.157 violation. Composes §11.4.26/.35/.17/.44/.65/.140/.156/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-157-PROPAGATION` (literal `11.4.157`, GEMINI.md INCLUDED) + recommended gate `CM-GEMINI-MD-LOCKSTEP` + paired §1.1 meta-test mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.157. Non-compliance is a release blocker. No escape hatch — no `--skip-gemini-md`, `--gemini-optional`, `--gemini-lag-OK`, `--four-carrier-suffices` flag.

§11.4.158 — Intensive all-feature/flow/edge-case video-recording + read-the-screen content-verification mandate (User mandate, 2026-06-16). Every project MUST be covered by intensive automated testing that exercises + RECORDS every feature/flow/use-case/edge-case (valid/invalid/boundary/concurrent/failure-injection §11.4.85), each recording showing the feature GENUINELY WORKING with REAL results (real prompts→real responses→real outputs §11.4.153) and NO false/simulated/stale/frozen result (§11.4.2/.5/.107) — a recording showing an error/bluff/non-working feature is a FINDING (→§11.4.153(4)/§11.4.4 fix→retest→re-record), never a confirmation. The testing System MUST ACTUALLY READ every shown log line / message / UI label / dialog / toast / status text and VERIFY it is a genuine

working result (OCR/ROI §11.4.117/.137 + confidence floor for pixel surfaces; direct capture for terminal/log; §11.4.107(10) self-validated golden-good/golden-bad analyzer) — "a video was produced" is NOT evidence, "the System read the screen + confirmed a genuine result" is. HelixQA (§11.4.27) MUST drive this exercise→record→read→score pass, PASS only on a read-confirmed genuine result + captured artefact path (§11.4.69). Default recording save path = `$HOME/Downloads` (host user's home Downloads, resolved at runtime never hardcoded) unless a project declares an override per §11.4.35; §11.4.155 project-prefix + §11.4.154 window-scope/rotation + §11.4.128 git-ignored raw corpus + §11.4.83 curated docs/qa evidence apply. Composes §11.4.2/.5/.25/.27/.52/.69/.83/.85/.107/.108/.117/.118/.128/.137/.138/.153/.154/.155/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-158-PROPAGATION` (literal `11.4.158`) + recommended gates `CM-INTENSIVE-RECORDING-COVERAGE` + `CM-RECORDING-CONTENT-READ-VERIFIED` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.158. Non-compliance is a release blocker. No escape hatch — no `--skip-recording-coverage`, `--video-without-content-read`, `--happy-path-recording-suffices`, `--recording-path-anywhere`, `--unread-recording-OK`, `--skip-helixqa-read` flag.